FACELESS CREATOR KIT

Space Facts Shorts Script Vault

FOR FACELESS SPACE & ASTRONOMY CHANNELS · SHORTS · REELS · TIKTOK

150 hooks · 60 timed scripts: 20× 30s · 20× 45s · 20× 60s

Verify with · Shelf-life tag · Myth check on every script

Evergreen · Re-check numbers

Every script names where to check it and flags which numbers can go out of date, so you don't post last year's moon count or a viral space myth.

PDF · 80 pages · Letter + A4
hooks.csv · scripts.csv · script_beats.csv

Faceless Creator Kit

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Every line is a link. Hooks are numbered H001–H150, scripts S01–S60. The small icon shows each script's Shelf-life tag (pine = Evergreen, circular arrow = Re-check numbers).

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S21–S40 · 85–112 words each

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60-second scripts

S41–S60 · 115–150 words each

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How to use this vault

150 space hooks and 60 finished scripts, each timed beat by beat: what to say, what to show, where to pause and how to sound. Seven steps from page to posted video:

1. **Pick a hook**
   Browse the hook library (pages 6–15): 10 hook types, 15 hooks each. Use one as an opener, a title idea or a cover line.

2. **Pick a script**
   Use the script index (page 16) to choose by length (30, 45 or 60 s), theme, topic and Shelf-life tag. Each script fits on one page.

3. **Check the "Verify with" line**
   Look the core claim up in the named reference before you record. The "Myth check" line shows the popular wrong version, if there is one.

4. **Re-check numbers if tagged**
   If the script is tagged Re-check numbers, confirm the figure named on its "Re-check" line in the named reference on the day you post. The words to update are highlighted.

5. **Swap the 3 "Make it yours" points**
   Each script lists 3 concrete swaps: a different opening, visual or question. Change at least one so the video is your own.

6. **Record, then add visuals per beat**
   Your own voice or any text-to-speech voice. Pause where the Caption / pause column says (‖ = pause, ▸ = new caption card). The Visual column names one visual type per beat.

7. **Post it**
   Copy the "Ready to post" box: title, one-line description, 3 hashtags, cover text, and the comment to pin.

The read-speed rule		2.5 words / second = 150 wpm
30 s 55–75 words | 45 s 85–112 | 60 s 115–150 · 2.5 words/s		
30 s scripts	5 beats	beat 1 = 0–3 s (the hook)
45 s scripts	6 beats	last beat = payoff + comment question
60 s scripts	8 beats	each beat: at most 2.5 × seconds + 2 words

Working in bulk?

The same text is in 3 CSV files: `hooks.csv`, `scripts.csv` (one row per script) and `script_beats.csv` (one row per beat). Import them into Google Sheets or Excel; see README.txt.

Accuracy, visuals & platform notes

Space channels get corrected in the comments fast, often about a myth or an out-of-date number. Every script carries one of these two Shelf-life tags:

Evergreen

Physics and long-settled facts that won't change, e.g. why the Moon shows one face.

Re-check numbers

Contains a count, record or mission status that can change, e.g. known moons or exoplanets. The figure is hedged (more than, about, as of the mid-2020s) and named on the Re-check line.

# RE-CHECK	**Number of confirmed moons of Saturn (and of Jupiter, if you compare them).** (S44; the voiceover says "well over two hundred")

We are not astronomers

The scripts are written from widely documented astronomy, and our build process checks them for internal consistency. **We are not astronomers, and no outside expert reviewed the scripts. Check each claim against the named reference before posting.**

If a viewer corrects you

Thank them, check the point in the named reference, and if they are right, pin a correction comment and fix the description.

Visual rights, illustrations and AI

Many NASA images are public domain, but agency logos and insignia are not free to use, and ESA and other agencies' images have their own licences: check each image's credit line. Label artist's impressions and AI images as illustrations, not real photos. Never use clips from films, TV shows or other creators. Follow each platform's AI-disclosure rules.

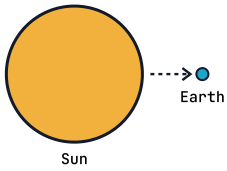
No promises

No script can promise views, followers, monetization or income. This vault saves research, writing and timing work; the channel is still yours to build.

01 OF 10

Scale It Down

Shrink something huge to a kitchen-table object so the size finally lands. Pair with a simple scale diagram in beat 2; say about when the ratio is rounded.




```
If [huge thing] were [everyday object], + [other thing] would
be [tiny object / distance]
```

H001	If the Sun were a basketball, Earth is a pinhead about 26 meters away.	Sun & stars	14 w
H002	Shrink Earth to a soccer ball, and Mars is a grapefruit.	Planets	11 w
H003	If Earth were a basketball, the Moon is a tennis ball seven meters away.	Moons	14 w
H004	Our Moon is wider than Pluto.	Moons	6 w
H005	About 1.3 million Earths could fit inside the Sun.	Sun & stars	9 w
H006	Jupiter is so big, all the other planets could fit inside it.	Planets	12 w
H007	Shrink the Milky Way to North America: the planets' orbits fit in a cup.	Galaxies & scale	14 w
H008	Sun as a sand grain? The nearest star would be about 30 kilometers away.	Galaxies & scale	14 w
H009	A black hole with the Sun's mass would be only a few kilometers across.	Black holes & extremes	14 w
H010	A teaspoon of neutron star would weigh about as much as a mountain.	Black holes & extremes	13 w
H011	On a desk globe, the space station would orbit about one centimeter up.	Humans in space	13 w
H012	If Earth were a marble, Saturn's rings would stretch wider than a dinner plate.	Planets	14 w
H013	The Sun is roughly 400 times wider than the Moon, and 400 times farther.	Sky myths	14 w
H014	A sand grain at arm's length covers a sky patch with thousands of galaxies.	Galaxies & scale	14 w
H015	Driving to the Moon at highway speed would take about five months nonstop.	Moons	13 w

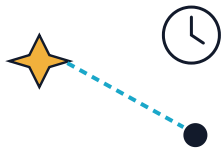
Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

02 OF 10

You’re Seeing the Past

Light takes time to travel, so everything in the sky is an old picture. Open on what the viewer sees, then reveal how old that light is.

`The [thing you see] + [left / happened] [time] ago`



H016	The sunlight on your skin left the Sun about eight minutes ago.	Sun & stars	12 w
H017	You see the Moon as it was about one second ago.	Moons	11 w
H018	Some stars you see tonight may have changed a lot since that light left.	Sun & stars	14 w
H019	The Andromeda Galaxy you see is light from about 2.5 million years ago.	Galaxies & scale	13 w
H020	Light from the Sun's core took ages to escape before its eight-minute trip.	Sun & stars	13 w
H021	Every photo of a distant galaxy is a picture of the past.	Galaxies & scale	12 w
H022	Look at Jupiter tonight: that light left it over half an hour ago.	Planets	13 w
H023	The farther you look into space, the further back in time you see.	Galaxies & scale	13 w
H024	The oldest light we can see is nearly 14 billion years old.	Galaxies & scale	12 w
H025	Radio signals from Voyager 1 take nearly a day to reach Earth.	Missions	12 w
H026	Mission control sees a Mars rover's moves minutes after they happen.	Missions	11 w
H027	Polaris looks steady, but its light is centuries old.	Sky myths	9 w
H028	The stars in Orion's belt shine with light from long before you were born.	Sky myths	14 w
H029	M87's black hole photo shows light that left about 55 million years ago.	Black holes & extremes	13 w
H030	Talking to the Moon, Apollo crews waited over two seconds for replies.	Humans in space	12 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

03 OF 10

How Long Would You Survive?

Drop the viewer somewhere hostile and ask how long they would last. Answer in beat 2 with the main danger, then explain why.

```
How long would you survive [place / situation]? + [the first thing that gets you]
```

H031	How long would you survive in space without a suit? Less than you think.	Humans in space	14 w
H032	Stand on Venus and you'd be crushed, baked and choked within moments.	Planets	12 w
H033	On Mars without a suit, you'd pass out in under a minute.	Planets	12 w
H034	Could you survive a night on the Moon? It lasts about two weeks.	Moons	13 w
H035	How long could you last on Jupiter? There's nowhere to stand.	Planets	11 w
H036	Falling into a black hole: how long before you'd notice anything?	Black holes & extremes	11 w
H037	On Titan, you'd need no pressure suit, just a mask and serious heating.	Moons	13 w
H038	How long would you last on Mercury's sunny side? Not long at all.	Planets	13 w
H039	Without a spacesuit, you'd have roughly 15 seconds of useful consciousness.	Humans in space	11 w
H040	Could you survive a dip in Europa's hidden ocean? First, the ice.	Moons	12 w
H041	Near a neutron star, the gravity alone would flatten you instantly.	Black holes & extremes	11 w
H042	How long can a human body last in zero gravity? Astronauts test it.	Humans in space	13 w
H043	Mars has air, but breathing it would knock you out within seconds.	Planets	12 w
H044	On Earth's highest peak you'd struggle to breathe. Space starts much higher.	Sky myths	12 w
H045	How long could you survive near the Sun? Spacecraft barely do.	Sun & stars	11 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

04 OF 10

Weird World

Lead with the strangest true detail about a planet, moon or exoplanet. Show it in beat 1, explain the cause in beat 2.

`[Place] has / does [strange fact]. + [why it happens]`



H046	Venus spins backward, so the Sun rises in the west there.	Planets	11 w
H047	Uranus rolls around the Sun on its side.	Planets	8 w
H048	Saturn's moon Enceladus sprays water ice into space.	Moons	8 w
H049	There's a moon with lakes, but they're filled with liquid methane.	Moons	11 w
H050	Io has hundreds of active volcanoes, and the heat comes from Jupiter.	Moons	12 w
H051	On Mars, sunsets can glow blue.	Planets	6 w
H052	Some exoplanets circle their star so fast that a year lasts under a day.	Galaxies & scale	14 w
H053	A day on Mercury lasts two of its years.	Planets	9 w
H054	Uranus's moon Miranda looks stitched together from mismatched pieces.	Moons	9 w
H055	Neptune's winds are some of the fastest in the solar system.	Planets	11 w
H056	Pluto has a giant heart-shaped plain made of frozen nitrogen.	Planets	10 w
H057	There are rogue planets drifting through space with no star at all.	Galaxies & scale	12 w
H058	Triton orbits Neptune backward, against the planet's spin.	Moons	8 w
H059	Saturn's moon Hyperion looks like a giant sponge.	Moons	8 w
H060	On some exoplanets, scientists think it may rain molten glass.	Galaxies & scale	10 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

05 OF 10

Look Up

Point the viewer at something they can see tonight without a telescope. No dates: say what to look for and why it looks that way.

`Tonight, look for [thing]. + [what it really is]`



H061	That bright dot sliding across the sky without blinking might be the space station.	Sky myths	14 w
H062	Meteor showers happen when Earth crosses old comet dust.	Sky myths	9 w
H063	You can see another galaxy with no telescope at all.	Galaxies & scale	10 w
H064	The Moon's dark part glows faintly sometimes. That's earthshine.	Moons	9 w
H065	Look for a star that doesn't twinkle. It's probably a planet.	Sky myths	11 w
H066	That pale band across a dark sky is our galaxy, seen from inside.	Galaxies & scale	13 w
H067	The evening star isn't a star at all. It's Venus.	Planets	10 w
H068	Follow Orion's belt down and you'll reach Sirius, the brightest star at night.	Sun & stars	13 w
H069	Watch the Moon for a week and its lit side keeps growing.	Moons	12 w
H070	In Orion, one bright star looks orange and another looks blue-white.	Sun & stars	11 w
H071	Shooting stars are mostly specks of dust, many no bigger than sand.	Sky myths	12 w
H072	The Moon looks huge on the horizon. Hold up your thumb to test it.	Moons	14 w
H073	On a dark night, satellites drift by as small moving dots.	Missions	11 w
H074	Polaris barely moves all night, while the stars around it circle.	Sky myths	11 w
H075	Jupiter's four big moons show up in ordinary binoculars.	Moons	9 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

06 OF 10

Space vs Earth

Compare one everyday Earth number with the same number somewhere else: a day, a year, your weight, a temperature.

`On Earth, [everyday number]. + On [place], [the same thing]`



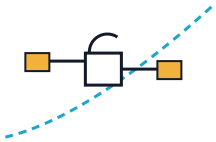
H076	A day on Venus is longer than a year on Venus.	Planets	11 w
H077	You'd weigh about a third as much on Mars.	Planets	9 w
H078	On the Moon, you'd weigh about one sixth of your Earth weight.	Moons	12 w
H079	A year on Neptune lasts about 165 Earth years.	Planets	9 w
H080	Earth spins once a day. Some dead stars spin hundreds of times a second.	Black holes & extremes	14 w
H081	Jupiter's day is under ten hours long.	Planets	7 w
H082	Mars has seasons like Earth, but each lasts about twice as long.	Planets	12 w
H083	The Sun's surface is hotter than any volcano on Earth, by thousands of degrees.	Sun & stars	14 w
H084	Earth's highest mountain would look small next to Olympus Mons on Mars.	Planets	12 w
H085	Astronauts on the station see about 16 sunrises a day.	Missions	10 w
H086	Earth is the densest planet in the solar system.	Planets	9 w
H087	A Moon day, sunrise to sunrise, lasts about a month on Earth.	Moons	12 w
H088	Saturn is far bigger than Earth but less dense than water.	Planets	11 w
H089	Astronauts can grow a few centimeters taller in orbit.	Humans in space	9 w
H090	A year on Mercury lasts about 88 Earth days.	Planets	9 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

07
OF 10

Mission Moment

Tell one documented moment from a spacecraft or mission, named by mission or spacecraft. No living people as the subject.



`[Mission / spacecraft] + [the moment that happened]`

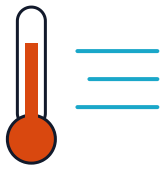
H091	Voyager 1 turned its camera back and caught Earth as a pale blue dot.	Missions	14 w
H092	Apollo 11's crew left footprints that may last for a very long time.	Missions	13 w
H093	Curiosity once played Happy Birthday on Mars, alone, to itself.	Missions	10 w
H094	Cassini ended its mission by diving into Saturn on purpose.	Missions	10 w
H095	New Horizons flew past Pluto and found a giant heart.	Planets	10 w
H096	Hubble launched with a flawed mirror, and astronauts fixed it in orbit.	Missions	12 w
H097	Rosetta's lander Philae bounced across a comet before settling in a shadow.	Missions	12 w
H098	The Huygens probe parachuted down to Titan and sent back pictures of its surface.	Moons	14 w
H099	The Apollo 15 crew drove a car on the Moon.	Missions	10 w
H100	Opportunity was planned for 90 days on Mars and lasted about 14 years.	Missions	13 w
H101	Sputnik 1 was a metal ball that beeped from orbit.	Missions	10 w
H102	Deep Impact smashed a probe into a comet on purpose.	Missions	10 w
H103	Pioneer 10 carries a plaque meant for anyone who might find it.	Missions	12 w
H104	Juno showed that Jupiter's poles are crowded with giant cyclones.	Planets	10 w
H105	Apollo 8's crew photographed Earth rising over the Moon.	Missions	9 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

08 OF 10

Cosmic Extremes

Biggest, hottest, fastest, densest. Say “known” or “we know of” with superlatives, and back the claim with one scale comparison.



`[The most extreme thing we know of] + [an everyday comparison]`

H106	The hottest planet isn't the closest one to the Sun.	Planets	10 w
H107	Magnetars have the strongest magnetic fields we know of.	Black holes & extremes	9 w
H108	Pulsars can keep time almost as steadily as atomic clocks.	Black holes & extremes	10 w
H109	Olympus Mons is the tallest volcano in the solar system that we know of.	Planets	14 w
H110	The Sun holds about 99.8 percent of the solar system's mass.	Sun & stars	11 w
H111	Space is cold, but the Sun's outer atmosphere reaches millions of degrees.	Sun & stars	12 w
H112	The Boomerang Nebula is colder than the Big Bang's leftover glow.	Galaxies & scale	11 w
H113	Jupiter's Great Red Spot is a storm wider than Earth.	Planets	10 w
H114	Some giant stars are wider than Earth's whole orbit around the Sun.	Sun & stars	12 w
H115	Black holes can weigh billions of times more than the Sun.	Black holes & extremes	11 w
H116	Ice giants like Neptune may rain diamonds deep inside.	Planets	9 w
H117	Some space probes have flown faster than any vehicle on Earth.	Missions	11 w
H118	Venus's surface is hot enough to melt lead.	Planets	8 w
H119	Light takes about 100,000 years to cross the Milky Way.	Galaxies & scale	10 w
H120	A day on Jupiter is the shortest of any planet.	Planets	10 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

09 OF 10

Myth Busted

Name the popular version first, then the documented one. Keep it friendly: the viewer heard it somewhere, they're not dumb.

`You've heard [space myth]. + Here's what's actually true.`



H121	You've heard the Moon has a dark side. Here's what's actually true.	Moons	12 w
H122	You've heard summer happens when Earth is closest to the Sun. It doesn't.	Sky myths	13 w
H123	You've heard the Great Wall is visible from space. Here's the truth.	Sky myths	12 w
H124	You've heard black holes suck everything in. Not quite.	Black holes & extremes	9 w
H125	You've heard the Sun is yellow. Here's its real color.	Sun & stars	10 w
H126	You've heard shooting stars are stars. They're not even close.	Sky myths	10 w
H127	You've heard there's no gravity in space. Here's what's really happening.	Humans in space	11 w
H128	You've heard Mercury is the hottest planet. It isn't.	Planets	9 w
H129	You've heard your blood would boil instantly in space. Here's the truth.	Humans in space	12 w
H130	You've heard a light-year measures time. It measures distance.	Galaxies & scale	9 w
H131	You've heard the North Star is the brightest star. It isn't.	Sky myths	11 w
H132	You've heard Earth's shadow causes Moon phases. Here's what really does.	Moons	11 w
H133	You've heard Jupiter has the most moons. That changed.	Moons	9 w
H134	You've heard eclipses give off harmful rays. Here's what's actually true.	Sky myths	11 w
H135	You've heard the Moon looks bigger near the horizon because it's closer. It isn't.	Moons	14 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

10 OF 10

Would You?

Ask a choice question in the first 3 seconds, give the facts that make it hard, and let the comments answer.



`Would you [space choice]? + [the catch]`

H136	Would you take a one-way trip to Mars?	Planets	8 w
H137	Would you live on the Moon for a year if you could?	Moons	12 w
H138	Would you spend six months on a space station with no showers?	Humans in space	12 w
H139	Would you eat freeze-dried food for a whole year?	Humans in space	9 w
H140	Would you rather see Saturn's rings up close or Jupiter's storms?	Planets	11 w
H141	Would you sleep strapped to a wall, floating?	Humans in space	8 w
H142	Would you watch a total eclipse from the Moon, or from Earth?	Sky myths	12 w
H143	Would you ride a rocket if the trip to orbit took nine minutes?	Humans in space	13 w
H144	Would you visit Titan, where it rains methane?	Moons	8 w
H145	Would you send your name to Mars on a microchip?	Missions	10 w
H146	Would you trade Earth's blue sky for Mars's blue sunsets?	Planets	10 w
H147	Would you live sealed inside a mock Mars base for a year?	Humans in space	12 w
H148	Would you rather live on a planet with two suns or none?	Galaxies & scale	12 w
H149	Would you stay up all night to count shooting stars?	Sky myths	10 w
H150	Would you go back to the Moon or push on to Mars?	Missions	12 w

Say it, then check it. Up to 9 words fit the 0–3 s hook beat; let a longer hook run into beat 2. Keep “about” on rounded numbers, and check the claim in the “Verify with” reference of the script you pair it with.

Script index

60 scripts: 20× 30s · 20× 45s · 20× 60s, across 8 themes. Each story is written at one length only, and each topic appears once. Click an ID or title to jump to it.

ID	TITLE	THEME	TOPIC	SHELF-LIFE	LENGTH	WORDS	PAGE
S01	A Day on Venus Is Longer Th...	Planets of the Solar Sys	Venus's rotation	Evergreen	30 s	73	18
S02	Why We Only See One Side o...	Moons & our Moon	Tidal locking	Evergreen	30 s	68	19
S03	The Sunlight on Your Skin Is 8...	Sun & stars	Sunlight travel ...	Evergreen	30 s	71	20
S04	Black Holes Can Turn You Int...	Black holes & extreme c	Spaghettification	Evergreen	30 s	70	21
S05	You Can See Another Galaxy ...	Galaxies & the scale of	Andromeda Gal...	Evergreen	30 s	66	22
S06	Voyager 1 Is Still Flying Away ...	Missions & exploration	Voyager 1	Re-check numb	30 s	65	23
S07	Astronauts Get Taller in Space	Humans in space (body	Spine growth i...	Evergreen	30 s	62	24
S08	Summer Isn't About Being Clo...	Sky myths & everyday a	Seasons	Evergreen	30 s	68	25
S09	The Tallest Volcano We Know...	Planets of the Solar Sys	Olympus Mons	Evergreen	30 s	68	26
S10	The Moon Is Slowly Drifting A...	Moons & our Moon	Lunar recession	Evergreen	30 s	72	27
S11	The Sun Isn't Yellow. It's White.	Sun & stars	The Sun's true c...	Evergreen	30 s	63	28
S12	A Teaspoon of Neutron Star ...	Black holes & extreme c	Neutron stars	Evergreen	30 s	66	29
S13	Hubble Stared at Nothing and...	Galaxies & the scale of	Hubble Deep Fi...	Evergreen	30 s	63	30
S14	Apollo 11's Footprints Are Pro...	Missions & exploration	Apollo 11 landi...	Evergreen	30 s	70	31
S15	Why Astronauts Eat Tortillas I...	Humans in space (body	Crumbs and sp...	Evergreen	30 s	65	32
S16	Shooting Stars Aren't Stars	Sky myths & everyday a	Meteors	Evergreen	30 s	64	33
S17	Uranus Spins on Its Side	Planets of the Solar Sys	Uranus's tilt	Evergreen	30 s	70	34
S18	Europa May Hide a Huge Oce...	Moons & our Moon	Europa's ocean	Re-check numb	30 s	68	35
S19	Betelgeuse Isn't About to Expl...	Sun & stars	Betelgeuse	Evergreen	30 s	67	36
S20	Black Holes Aren't Cosmic Va...	Black holes & extreme c	Black hole grav...	Evergreen	30 s	64	37
S21	A Light-Year Is a Distance, No...	Galaxies & the scale of	Light-year	Evergreen	45 s	102	38
S22	The Mars Rover That Sang Ha...	Missions & exploration	Curiosity rover	Re-check numb	45 s	94	39
S23	How Astronauts Sleep With N...	Humans in space (body	Sleeping in orbit	Evergreen	45 s	101	40
S24	Why the Sky Is Blue (It's Not t...	Sky myths & everyday a	Why the sky is ...	Evergreen	45 s	98	41
S25	Saturn Is Less Dense Than W...	Planets of the Solar Sys	Saturn's density	Evergreen	45 s	94	42
S26	Earth's Shadow Doesn't Caus...	Moons & our Moon	Lunar phases	Evergreen	45 s	96	43
S27	Sunlight's Energy May Be Old...	Sun & stars	Energy escapin...	Evergreen	45 s	98	44
S28	The First Image of a Black Hole	Black holes & extreme c	M87 black hole ...	Evergreen	45 s	100	45
S29	Earth Is Speeding Around the ...	Galaxies & the scale of	Galactic year	Evergreen	45 s	103	46
S30	Hubble Launched Blurry and ...	Missions & exploration	Hubble's mirro...	Re-check numb	45 s	90	47

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S32	Solar Eclipses Rely on a Cos...	Sky myths & everyday a	Total solar ecli...	Evergreen	45 s	96	49
S33	Sunsets on Mars Are Blue	Planets of the Solar Sys	Blue Martian su...	Evergreen	45 s	91	50
S34	The Giant Rising Moon Is an Ill...	Moons & our Moon	Moon illusion	Evergreen	45 s	96	51
S35	Our Nearest Star Neighbor Is ...	Sun & stars	Proxima Centa...	Re-check numb	45 s	97	52
S36	Magnetars: Dead Stars That A...	Black holes & extreme d	Magnetars	Evergreen	45 s	98	53
S37	We Can See Farther Than 13....	Galaxies & the scale of	Observable uni...	Evergreen	45 s	99	54
S38	New Horizons Found a Giant ...	Missions & exploration	New Horizons a...	Evergreen	45 s	94	55
S39	Why Astronauts Work Out Tw...	Humans in space (body	Bone and musc...	Evergreen	45 s	93	56
S40	The North Star Isn't the Bright...	Sky myths & everyday a	Polaris	Evergreen	45 s	95	57
S41	Jupiter's Great Red Spot Is Sh...	Planets of the Solar Sys	Great Red Spot	Re-check numb	60 s	120	58
S42	What Really Happens to a Bo...	Humans in space (body	Vacuum expos...	Evergreen	60 s	122	59
S43	The Sun Has an Eleven-Year ...	Sun & stars	Solar cycle	Evergreen	60 s	128	60
S44	Saturn's Moon Count Keeps ...	Moons & our Moon	Moons of Saturn	Re-check numb	60 s	125	61
S45	The Space Signal Nicknamed ...	Black holes & extreme d	Pulsars	Evergreen	60 s	122	62
S46	We've Found Thousands of Pl...	Galaxies & the scale of	Exoplanet count	Re-check numb	60 s	125	63
S47	Cassini's Final Dive Into Saturn	Missions & exploration	Cassini Grand F...	Evergreen	60 s	129	64
S48	Space Is Closer Than You Think	Humans in space (body	People who hav...	Re-check numb	60 s	131	65
S49	You Can't See the Great Wall ...	Sky myths & everyday a	Great Wall fro...	Evergreen	60 s	134	66
S50	The Closest Planet to the Sun ...	Planets of the Solar Sys	Mercury's temp...	Evergreen	60 s	128	67
S51	Io: The Moon Covered in Volc...	Moons & our Moon	Io's volcanoes	Evergreen	60 s	135	68
S52	Why the Sun's Atmosphere Is ...	Sun & stars	Coronal heating	Re-check numb	60 s	120	69
S53	The Giant Black Hole at the H...	Black holes & extreme d	Sagittarius A*	Evergreen	60 s	131	70
S54	The Oldest Light in the Univer...	Galaxies & the scale of	Cosmic microw...	Evergreen	60 s	134	71
S55	People Have Lived in Orbit No...	Missions & exploration	International S...	Re-check numb	60 s	129	72
S56	The Longest Single Trip to Sp...	Humans in space (body	Longest single s...	Re-check numb	60 s	133	73
S57	Why Stars Twinkle but Planet...	Sky myths & everyday a	Twinkling	Evergreen	60 s	132	74
S58	Neptune Was Found With Math	Planets of the Solar Sys	Neptune's disco...	Evergreen	60 s	136	75
S59	Our Moon May Have Formed ...	Moons & our Moon	Giant-impact h...	Evergreen	60 s	136	76
S60	Auroras Happen at Both Ends ...	Sky myths & everyday a	Auroras	Evergreen	60 s	132	77

A Day on Venus Is Longer Than Its Year

BEAT TIMELINE (SECONDS) · TOPIC: VENUS'S ROTATION

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	On Venus, one day lasts longer than one year.	Labelled artist's impression of Venus, slowly turning	▸ "DAY > YEAR"	Curious
3–9	Venus takes about 243 Earth days to spin once. But it circles the Sun in about 225.	Simple diagram: spin arrow vs orbit path, two counters	▸ New card per number	Clear
9–16	And it spins backward compared to most planets, so the Sun rises in the west there.	Animated icons: Sun rising over a Venus horizon, west label	‖ Pause after "backward"	Surprised
16–23	Its sunrise-to-sunrise day is shorter, about 117 Earth days, because the spin and orbit work against each other.	Scale diagram: three bars, 243, 225 and 117 days	Highlight "117 Earth days"	Explaining
23–30 CTA	Would you rather have a longer day or a longer year? Comment below.	Venus icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Venus Facts · Encyclopaedia Britannica: Venus (planet)
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a calendar flipping while Venus barely turns.
2. Compare with Mercury, where a solar day lasts two of its years.
3. Ask viewers to guess the day length before the reveal.

Ready to post

TITLE A Day on Venus Is Longer Than Its Year

DESCRIPTION Venus spins so slowly that its day outlasts its year.

HASHTAGS #space #venus #planets

COVER TEXT Day longer than a year?

PIN Want the Mercury version next?

Why We Only See One Side of the Moon

BEAT TIMELINE (SECONDS) · TOPIC: TIDAL LOCKING

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	We only ever see one side of the Moon.	Public-domain agency image of the full Moon, credit shown	▸ "ONE SIDE ONLY"	Curious
3–9	The Moon spins once per trip around Earth, so the same face always points at us.	Animated diagram: Moon orbiting Earth, one face marked	▸ New card per clause	Clear
9–16	Earth's gravity slowed its spin long ago until it locked. Scientists call this tidal locking.	Simple diagram: slowing spin arrows, then lock icon	Highlight "tidal locking"	Explaining
16–23	But the far side isn't dark. It gets just as much sunlight, just never facing us.	Animated icons: Sun lighting the far side	‖ Pause after "isn't dark"	Firm
23–30 CTA	Did you think the far side was always dark? Be honest below.	Moon icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Moon Facts · Encyclopaedia Britannica: Moon, rotation section
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** The far side of the Moon is always dark. **Documented:** It gets about as much sunlight as the near side; we just never see it from Earth.

Make it yours

3 swaps so it's your video

1. Open with a real-time spin demo: a ball circling a lamp.
2. Mention that Pluto and Charon are locked to each other.
3. Ask viewers what they think is on the far side.

Ready to post

TITLE Why We Only See One Side of the Moon

DESCRIPTION The Moon turns, just at the perfect speed to hide one side.

HASHTAGS #space #moon #astronomy

COVER TEXT The Moon's hidden side

PIN Should I cover the far side's craters next?

The Sunlight on Your Skin Is 8 Minutes Old

BEAT TIMELINE (SECONDS) · TOPIC: SUNLIGHT TRAVEL TIME

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	This sunlight left the Sun about eight minutes ago.	Stock clip of sunlight through a window	▸ "8 MINUTES OLD"	Calm
3–9	Light is the fastest thing there is, but the Sun is about 150 million kilometers away.	Scale diagram: Sun, Earth, distance arrow labelled	Highlight "150 million km"	Clear
9–16	So if the Sun suddenly vanished, you'd keep seeing it for about eight more minutes.	Animated icons: Sun fading after a timer	‖ Pause after "vanished"	Dramatic
16–23	Moonlight is even fresher. It's sunlight that bounced off the Moon about a second before reaching you.	Simple diagram: light path Sun to Moon to eye	▸ New card per sentence	Light
23–30 CTA	Which old light do you want to see next, Mars or a distant star?	Mars icon, star icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Sun Facts · Encyclopaedia Britannica: light-year and speed of light entries
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a stopwatch counting to eight minutes over a sunrise.
2. Swap Moonlight for light from Mars or Jupiter in beat 4.
3. Ask viewers to guess the travel time first.

Ready to post

TITLE The Sunlight on Your Skin Is 8 Minutes Old

DESCRIPTION Light is fast, but the Sun is very far away.

HASHTAGS #space #sun #science

COVER TEXT This sunlight is old

PIN Which planet's light should I time next?

Black Holes Can Turn You Into Spaghetti

BEAT TIMELINE (SECONDS) · TOPIC: SPAGHETTIFICATION

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Near some black holes, you'd be stretched like spaghetti.	Labelled artist's impression of a black hole	▸ "SPAGHETTI?"	Playful
3–9	Gravity pulls your feet harder than your head. Near a black hole, that gap gets huge.	Simple diagram: stick figure, two arrows of different length	▸ New card per sentence	Clear
9–16	Your body would be stretched long and squeezed thin. Scientists really call it spaghettification.	Animated icons: figure stretching into a noodle	Highlight "spaghettification"	Amused
16–23	Strangely, a giant black hole is gentler at its edge. You could cross it before the stretching gets bad.	Scale diagram: small vs giant black hole	‖ Pause after "gentler"	Surprised
23–30 CTA	Small black hole or giant one: which would you rather fall into?	Two black hole icons + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Black Holes · Encyclopaedia Britannica: black hole
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Use a rubber band stretching animation as the hook visual.
2. Explain tides on Earth as the gentle version of the same effect.
3. Ask viewers to name the scariest black hole fact they know.

Ready to post

TITLE Black Holes Can Turn You Into Spaghetti

DESCRIPTION Why falling into a small black hole stretches you thin.

HASHTAGS #space #blackhole #physics

COVER TEXT Stretched like spaghetti

PIN Want a script on giant black holes next?

You Can See Another Galaxy Without a Telescope

BEAT TIMELINE (SECONDS) · TOPIC: ANDROMEDA GALAXY

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	You can see another galaxy with just your eyes.	Stock clip of a dark night sky	▸ "ANOTHER GALAXY"	Wonder
3–9	On a dark night, the Andromeda Galaxy looks like a faint, fuzzy smudge.	Simple sky chart with the smudge circled	Highlight "fuzzy smudge"	Calm
9–16	That smudge holds hundreds of billions of stars, and its light left about 2.5 million years ago.	Public-domain agency image of Andromeda, credit shown	‖ Pause after "stars"	Building
16–23	It's one of the most distant things most people can spot without a telescope or binoculars.	Scale diagram: Milky Way and Andromeda with distance arrow	▸ New card per clause	Clear
23–30 CTA	Have you ever seen Andromeda yourself? Tell me where you were.	Andromeda icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Andromeda Galaxy (Messier 31) · Encyclopaedia Britannica: Andromeda Galaxy
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Show a simple star-hop map from a bright star you know.
2. Compare its true width on the sky with the full Moon.
3. Ask viewers how dark their sky is at home.

Ready to post

TITLE You Can See Another Galaxy Without a Telescope

DESCRIPTION That faint smudge is a galaxy of hundreds of billions of stars.

HASHTAGS #space #galaxy #stargazing

COVER TEXT A galaxy, no telescope

PIN Want a star-hop guide to find it?

Voyager 1 Is Still Flying Away From Us

BEAT TIMELINE (SECONDS) · TOPIC: VOYAGER 1

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Voyager 1 has been flying away since 1977.	Labelled artist's impression of a deep-space probe	▸ "SINCE 1977"	Curious
3–9	As of the mid-2020s, it's **more than 24 billion kilometers** from Earth.	Scale diagram: Sun, planets, probe far to the right	Highlight "24 billion km"	Clear
9–16	That makes it the farthest spacecraft humans have built, and it's still sending faint signals home.	Animated icons: probe sending signal waves to Earth	‖ Pause after "built"	Proud
16–23	Its signals travel at the speed of light and still take most of a day to arrive.	Simple diagram: signal path with a clock icon	▸ New card per clause	Amazed
23–30 CTA	What message would you send on a probe leaving the solar system?	Probe icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA JPL: Voyager mission status page · Encyclopaedia Britannica: Voyager (space probes)
SHELF-LIFE	**Re-check numbers.** Distance grows every day and the probe's power is running down; the voiceover hedges both.
# RE-CHECK	**Voyager 1's current distance from Earth and whether it still sends data.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on the launch year 1977 and a counter running upward.
2. Add how long its radio signal takes to arrive today.
3. Ask what viewers would put on a message to aliens.

Ready to post

TITLE Voyager 1 Is Still Flying Away From Us

DESCRIPTION A 1977 probe is still heading out of the solar system.

HASHTAGS #space #voyager #nasa

COVER TEXT Launched 1977. Still going.

PIN Should I cover Voyager 2 next?

Astronauts Get Taller in Space

BEAT TIMELINE (SECONDS) · TOPIC: SPINE GROWTH IN ORBIT

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Astronauts can get taller in space.	Animated icons: height chart, mark moving up	▸ "TALLER IN SPACE"	Curious
3–9	On Earth, gravity squeezes the soft discs in your spine all day long.	Simple diagram: spine with discs pressed down	Highlight "discs"	Clear
9–16	In orbit, those discs can spread out, and astronauts can gain a few centimeters of height.	Simple diagram: same spine, discs spread apart	‖ Pause after "spread out"	Building
16–23	Some report back pain from the stretch, so suits and seats need extra room.	Animated icons: back with pain marks, seat outline	▸ New card per clause	Serious
23–30 CTA	Back on Earth, gravity squeezes them back. Would you like the extra height?	Height chart icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA: Human Research Program, spine and back pain in spaceflight · Encyclopaedia Britannica: weightlessness
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a height chart with a pencil mark moving up.
2. Mention that you are slightly taller in the morning too.
3. Ask viewers how tall they would like to be.

Ready to post

TITLE Astronauts Get Taller in Space

DESCRIPTION Without gravity squeezing the spine, astronauts stretch out a little.

HASHTAGS #space #astronaut #science

COVER TEXT Taller in space?

PIN Want more on what space does to your body?

Summer Isn’t About Being Closer to the Sun

BEAT TIMELINE (SECONDS) · TOPIC: SEASONS

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Summer isn't caused by being closer to the Sun.	Stock clip of a sunny beach	▸ “NOT DISTANCE”	Firm
3–9	Earth is actually closest to the Sun in early January, during northern winter.	Simple diagram: Earth's orbit, January position marked	Highlight “early January”	Surprised
9–16	Seasons come from Earth's tilt of about 23 degrees. Your half leans toward or away from the Sun.	Animated diagram: tilted Earth orbiting the Sun	▸ New card per sentence	Explaining
16–23	Leaning toward it means higher sunlight and longer days. That's summer, while the other half has winter.	Simple diagram: two hemispheres, summer and winter labels	‖ Pause after “summer”	Clear
23–30 CTA	Did you learn the distance version at school? Tell me below.	Globe icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Space Place: What Causes the Seasons? · Encyclopaedia Britannica: season
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Summer comes when Earth is closest to the Sun. **Documented:** Seasons come from Earth's tilt; Earth is closest to the Sun in early January.

Make it yours

3 swaps so it’s your video

1. Open with a tilted globe and a flashlight on a table.
2. Point out that Australia has summer in December.
3. Ask viewers which season they love most.

Ready to post

TITLE Summer Isn't About Being Closer to the Sun

DESCRIPTION The tilt of Earth, not its distance, makes the seasons.

HASHTAGS #space #seasons #earth

COVER TEXT Closer means summer? No.

PIN Who learned the distance version at school?

The Tallest Volcano We Know Is on Mars

BEAT TIMELINE (SECONDS) · TOPIC: OLYMPUS MONS

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The tallest volcano we know of is on Mars.	Labelled artist's impression of Olympus Mons	▸ "TALLEST VOLCANO"	Curious
3–9	Olympus Mons rises about 22 kilometers, roughly two and a half times Mount Everest.	Scale diagram: Everest beside Olympus Mons	Highlight "22 km"	Clear
9–16	Its base is so wide that, standing on it, the edges would sink below your horizon.	Simple diagram: curved horizon hiding the volcano's edges	‖ Pause after "so wide"	Wonder
16–23	It grew so huge because Mars has no moving plates, so lava kept piling up in one spot.	Animated icons: lava layers stacking over one hotspot	▸ New card per clause	Explaining
23–30 CTA	Would you climb a volcano taller than Everest if you could?	Mountain icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Mars Facts · Encyclopaedia Britannica: Olympus Mons
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on Mount Everest, then zoom out to Olympus Mons.
2. Compare its base with the size of a country you know.
3. Ask viewers which mountain they would climb first.

Ready to post

TITLE The Tallest Volcano We Know Is on Mars

DESCRIPTION Olympus Mons is about two and a half times Everest's height.

HASHTAGS #space #mars #volcano

COVER TEXT 2.5 Everests tall

PIN Want the biggest canyon on Mars next?

The Moon Is Slowly Drifting Away

BEAT TIMELINE (SECONDS) · TOPIC: LUNAR RECESSION

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The Moon is slowly drifting away from Earth.	Public-domain agency image of the Moon, credit shown	▸ "DRIFTING AWAY"	Curious
3–9	Every year it moves about 3.8 centimeters farther, roughly as fast as your fingernails grow.	Scale diagram: ruler showing 3.8 cm	Highlight "3.8 cm"	Clear
9–16	Scientists measure this by bouncing lasers off mirrors that Apollo astronauts left on the surface.	Animated icons: laser beam from Earth to Moon and back	‖ Pause after "lasers"	Building
16–23	The cause is tides. The Moon pulls on Earth's oceans, and that tug slowly pushes the Moon outward.	Simple diagram: ocean bulge pulling the Moon forward	▸ New card per sentence	Explaining
23–30 CTA	It's far too slow to notice in a lifetime. Did you know the Moon was leaving?	Moon icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Moon Facts · Encyclopaedia Britannica: Moon, tidal evolution section
SHELF-LIFE	Evergreen. A slow, long-measured average; the yearly figure is stable for any channel's lifetime.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open with a fingernail growth time-lapse as the comparison.
2. Explain the laser mirrors with a simple flashlight demo.
3. Ask viewers to guess the yearly drift first.

Ready to post

TITLE The Moon Is Slowly Drifting Away

DESCRIPTION The Moon moves a few centimeters farther from Earth every year.

HASHTAGS #space #moon #science

COVER TEXT The Moon is leaving

PIN Want to know how far it was long ago?

The Sun Isn’t Yellow. It’s White.

BEAT TIMELINE (SECONDS) · TOPIC: THE SUN'S TRUE COLOR

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The Sun isn't really yellow. It's white.	Simple graphic: yellow Sun icon turning white	▸ “NOT YELLOW”	Surprised
3–9	It gives off light in every color, and together those colors look white to our eyes.	Animated icons: rainbow bands merging into white	▸ New card per clause	Clear
9–16	Astronauts in orbit see a white Sun, because no air sits in the way.	Labelled artist's impression of the Sun above Earth's edge	Highlight “no air”	Calm
16–23	From the ground, air scatters some blue light, so the Sun looks yellow or orange.	Simple diagram: sunlight passing through the atmosphere	‖ Pause after “blue light”	Explaining
23–30 CTA	Did you grow up drawing the Sun yellow? Tell me below.	Crayon Sun icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Sun Facts · Encyclopaedia Britannica: Sun, and Rayleigh scattering
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** The Sun is a yellow star. **Documented:** Its light is white; air scattering makes it look yellow or orange from the ground.

Make it yours

3 swaps so it’s your video

1. Open with a prism splitting white light into colors.
2. Link it to why sunsets look orange.
3. Ask viewers what color they always drew the Sun.

Ready to post

TITLE The Sun Isn't Yellow. It's White.

DESCRIPTION Seen from space, the Sun's light is white, not yellow.

HASHTAGS #space #sun #science

COVER TEXT The Sun is white?

PIN Should I explain red sunsets next?

A Teaspoon of Neutron Star Weighs a Mountain

BEAT TIMELINE (SECONDS) · TOPIC: NEUTRON STARS

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	One teaspoon of neutron star weighs about a mountain.	Animated icons: teaspoon and mountain on a balance scale	▸ "1 TEASPOON = 1 MOUNTAIN"	Dramatic
3–9	A neutron star is what's left when a giant star collapses after a supernova.	Labelled artist's impression of a supernova	‖ Pause after "left"	Clear
9–16	It packs more mass than the Sun into a ball about 20 kilometers across, roughly a city.	Scale diagram: neutron star over a city outline	Highlight "20 km"	Amazed
16–23	The matter is crushed so hard that atoms collapse, leaving mostly neutrons packed together.	Simple diagram: atom shrinking into packed dots	▸ New card per clause	Explaining
23–30 CTA	What would you compare a spoonful of neutron star to? Comment below.	Teaspoon icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Neutron Stars · Encyclopaedia Britannica: neutron star
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a teaspoon next to a mountain silhouette.
2. Compare the 20 km width with a city your viewers know.
3. Ask viewers what object they would compare it to.

Ready to post

TITLE A Teaspoon of Neutron Star Weighs a Mountain

DESCRIPTION More than the Sun's mass, crushed to the size of a city.

HASHTAGS #space #neutronstar #physics

COVER TEXT A spoon that weighs a mountain

PIN Want a script on pulsars next?

Hubble Stared at Nothing and Found Galaxies

BEAT TIMELINE (SECONDS) · TOPIC: HUBBLE DEEP FIELD

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Hubble stared at an empty patch of sky.	Black square with a faint frame, slow zoom	▸ "EMPTY SKY?"	Curious
3–9	In 1995, it pointed at a tiny, dark spot for about ten days straight.	Animated icons: telescope icon with a 10-day counter	Highlight "ten days"	Clear
9–16	The patch was tiny, like a tennis ball seen from about 100 meters away.	Scale diagram: tennis ball at the end of a field	‖ Pause after "tiny"	Building
16–23	That empty spot was full of galaxies, about three thousand of them, each with countless stars.	Public-domain agency image of the Hubble Deep Field, credit shown	▸ "~3,000 GALAXIES"	Wonder
23–30 CTA	Does that make you feel big or tiny? Tell me below.	Galaxy icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science / ESA Hubble: Hubble Deep Field · Encyclopaedia Britannica: Hubble Deep Field
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a black square, then fade in the galaxies.
2. Mention the later, even deeper Ultra Deep Field image.
3. Ask viewers to guess how many galaxies were found.

Ready to post

TITLE Hubble Stared at Nothing and Found Galaxies

DESCRIPTION A tiny dark patch of sky held thousands of galaxies.

HASHTAGS #space #hubble #galaxies

COVER TEXT Empty sky? Not empty.

PIN Want the Ultra Deep Field version next?

Apollo 11's Footprints Are Probably Still There

BEAT TIMELINE (SECONDS) · TOPIC: APOLLO 11 LANDING SITE

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Apollo 11's footprints are probably still on the Moon.	Public-domain agency image of a lunar bootprint, credit shown	▸ "STILL THERE?"	Curious
3–9	The Moon has no wind or rain, so nothing on its surface gets washed or blown away.	Animated icons: wind and rain icons crossed out	▸ New card per clause	Clear
9–16	Tiny meteorite impacts and harsh sunlight slowly change the dust, but over a very long time.	Simple diagram: tiny dots hitting dust, slow clock	‖ Pause after "dust"	Calm
16–23	The flag never waved in wind. A rod held it out, and it moved only when handled.	Simple diagram: flag with a horizontal top rod labelled	Highlight "a rod held it out"	Firm
23–30 CTA	Would you visit the Apollo 11 site or keep it untouched?	Footprint icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA: Apollo 11 mission overview · Encyclopaedia Britannica: Apollo 11
SHELF-LIFE	**Evergreen.** A historical mission; nothing here changes year to year.
MYTH CHECK	**Popular version:** The flag waving shows the landing was filmed on Earth. **Documented:** A horizontal rod held the flag out; it moved only while being handled, with no air to stop it.

Make it yours

3 swaps so it's your video

1. Open on a bootprint photo described as public domain, credit shown.
2. Compare with how fast footprints vanish on a beach.
3. Ask viewers whether the site should be protected.

Ready to post

TITLE Apollo 11's Footprints Are Probably Still There

DESCRIPTION With no wind or rain, the Moon keeps footprints a long time.

HASHTAGS #space #apollo11 #moon

COVER TEXT Still on the Moon

PIN Should landing sites be protected like parks?

Why Astronauts Eat Tortillas Instead of Bread

BEAT TIMELINE (SECONDS) · TOPIC: CRUMBS AND SPACE FOOD

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Bread crumbs are a real problem in space.	Stock clip of bread crumbling, slowed down	▸ "CRUMBS = PROBLEM"	Playful
3–9	In microgravity, crumbs don't fall. They float into air vents, equipment and even astronauts' eyes.	Animated icons: crumbs drifting toward a vent and an eye	▸ New card per sentence	Clear
9–16	That's one reason crews often use tortillas instead of bread. They make far fewer crumbs.	Simple graphic: bread crossed out, tortilla with a tick	Highlight "tortillas"	Light
16–23	Salt and pepper come as liquids, because loose grains would drift off and clog things.	Animated icons: two small dropper bottles	‖ Pause after "liquids"	Amused
23–30 CTA	Which food would you miss most on a space station? Comment below.	Fork icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA: Space Food Systems · Encyclopaedia Britannica: space food
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a slice of bread crumbling in slow motion.
2. Show a tortilla wrap being folded as the payoff visual.
3. Ask viewers what snack they would bring to orbit.

Ready to post

TITLE Why Astronauts Eat Tortillas Instead of Bread

DESCRIPTION In microgravity, every crumb floats, so menus are planned around it.

HASHTAGS #space #astronaut #food

COVER TEXT No bread in space?

PIN Want to know how astronauts drink coffee?

Shooting Stars Aren’t Stars

BEAT TIMELINE (SECONDS) · TOPIC: METEORS

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Shooting stars aren't stars at all.	Stock clip of a night sky with a streak graphic	▸ “NOT STARS”	Curious
3–9	They're bits of space dust and rock burning up in our air, high above the ground.	Simple diagram: atmosphere layers, glowing streak up high	▸ New card per clause	Clear
9–16	Many are no bigger than a grain of sand, but they hit the air incredibly fast.	Scale diagram: sand grain beside a speed arrow	Highlight “grain of sand”	Surprised
16–23	That speed heats the air around them until it glows, which is the streak you see.	Animated icons: speck with a glowing trail	‖ Pause after “glows”	Explaining
23–30 CTA	What did you wish for on your last shooting star?	Star icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Meteors and Meteorites · Encyclopaedia Britannica: meteor
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Shooting stars are stars falling from the sky. **Documented:** They're specks of dust or rock burning up in Earth's upper atmosphere.

Make it yours

3 swaps so it’s your video

1. Open on a long-exposure style streak graphic.
2. Explain meteor showers as Earth crossing comet dust.
3. Ask viewers where they saw their best shooting star.

Ready to post

TITLE Shooting Stars Aren't Stars

DESCRIPTION That streak is a speck of dust burning up high above you.

HASHTAGS #space #meteor #stargazing

COVER TEXT Not a star at all

PIN Want a meteor shower guide next?

Uranus Spins on Its Side

BEAT TIMELINE (SECONDS) · TOPIC: URANUS'S TILT

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Uranus spins tipped over on its side.	Labelled artist's impression of Uranus with faint rings	▸ "ON ITS SIDE"	Curious
3–9	Its tilt is about 98 degrees, so its poles take turns pointing almost straight at the Sun.	Simple diagram: tilted axis at 98 degrees	Highlight "98 degrees"	Clear
9–16	One Uranus year lasts about 84 Earth years, so each pole gets roughly 42 years of sunlight.	Animated diagram: Uranus orbit, sunlit pole switching	▸ New card per clause	Building
16–23	Then about 42 years of darkness. A leading idea is that a huge collision knocked it over long ago.	Animated icons: object striking a planet, planet tipping	‖ Pause after "darkness"	Dramatic
23–30 CTA	Would you live through a 42-year summer? Tell me below.	Uranus icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Uranus Facts · Encyclopaedia Britannica: Uranus (planet)
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a spinning top falling onto its side.
2. Compare with Earth's tilt of about 23 degrees.
3. Ask viewers if they would pick 42 years of day or night.

Ready to post

TITLE Uranus Spins on Its Side

DESCRIPTION Uranus is tipped about 98 degrees, so its seasons are wild.

HASHTAGS #space #uranus #planets

COVER TEXT A planet on its side

PIN Want the Venus backward-spin script too?

Europa May Hide a Huge Ocean Under Ice

BEAT TIMELINE (SECONDS) · TOPIC: EUROPA'S OCEAN

0 · 3 · 9 · 16 · 23 · 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Jupiter's moon Europa probably hides an ocean.	Labelled artist's impression of Europa's cracked ice	▸ "A HIDDEN OCEAN"	Curious
3–9	Under its ice, scientists think a salty ocean holds more water than Earth's oceans combined.	Simple diagram: ice shell over a deep ocean layer	Highlight "more water"	Amazed
9–16	Jupiter's gravity squeezes Europa as it orbits, and that flexing makes enough heat to keep water liquid.	Animated icons: moon squeezed and stretched by Jupiter	‖ Pause after "squeezes"	Explaining
16–23	A spacecraft called Europa Clipper launched in 2024, and **as of the mid-2020s it's on its way there**.	Animated icons: generic probe on a dashed path to Jupiter	▸ New card per clause	Building
23–30 CTA	Do you think anything lives in Europa's ocean? Tell me why.	Europa icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Europa · NASA: Europa Clipper mission page · Encyclopaedia Britannica: Europa (moon)
SHELF-LIFE	**Re-check numbers.** Mission status changes; the voiceover dates it to the mid-2020s.
# RE-CHECK	**Europa Clipper's current mission status and arrival plans.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a cracked ice texture with water glowing underneath.
2. Compare the ice shell with the deepest ocean on Earth.
3. Ask viewers what they hope the probe finds.

Ready to post

TITLE Europa May Hide a Huge Ocean Under Ice

DESCRIPTION Jupiter's moon Europa likely has a salty ocean under its ice.

HASHTAGS #space #europa #jupiter

COVER TEXT An ocean under the ice

PIN Should I follow Europa Clipper's progress?

Betelgeuse Isn't About to Explode Tomorrow

BEAT TIMELINE (SECONDS) · TOPIC: BETELGEUSE

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Betelgeuse is a giant red star in Orion.	Simple sky chart of Orion, Betelgeuse circled	▸ "BETELGEUSE"	Curious
3–9	Put it where our Sun is, and it could reach past the orbit of Mars.	Scale diagram: giant star circle over planet orbits	Highlight "past Mars"	Amazed
9–16	Around 2020 it dimmed so much that people asked if it was about to explode.	Animated icons: star icon fading, question marks	‖ Pause after "dimmed"	Building
16–23	Studies mostly blamed a dust cloud. It will explode someday, probably not for a very long time.	Labelled artist's impression of a dust cloud near a star	▸ New card per sentence	Calm
23–30 CTA	If it did explode, would you stay up to watch? Comment below.	Star icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Betelgeuse's Great Dimming (Hubble) · Encyclopaedia Britannica: Betelgeuse
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Betelgeuse is about to explode any day now. **Documented:** Its big dimming around 2020 was mostly dust; a supernova is expected eventually, likely not soon.

Make it yours

3 swaps so it's your video

1. Open on Orion with Betelgeuse circled in red.
2. Compare its size with the orbits of Earth and Mars.
3. Ask viewers if they have spotted Betelgeuse themselves.

Ready to post

TITLE Betelgeuse Isn't About to Explode Tomorrow

DESCRIPTION Orion's red giant is huge, but its famous dimming was mostly dust.

HASHTAGS #space #betelgeuse #stars

COVER TEXT Is Betelgeuse exploding?

PIN Want a script on what a supernova is?

Black Holes Aren’t Cosmic Vacuum Cleaners

BEAT TIMELINE (SECONDS) · TOPIC: BLACK HOLE GRAVITY

0 3 9 16 23 30

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Black holes are not cosmic vacuum cleaners.	Simple graphic: vacuum cleaner icon crossed out	▸ “NOT A VACUUM”	Firm
3–9	From far away, a black hole pulls just like any other object with the same mass.	Simple diagram: two equal masses, equal gravity arrows	▸ New card per clause	Clear
9–16	If the Sun became a black hole of the same mass, Earth's orbit wouldn't change.	Animated diagram: Sun swapped for a dot, orbit unchanged	Highlight “wouldn't change”	Surprised
16–23	It would get dark and cold, but nothing gets pulled in unless it comes very close.	Animated icons: Earth cooling, orbit still steady	‖ Pause after “cold”	Explaining
23–30 CTA	Did you think black holes suck everything in? Be honest.	Black hole icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Black Holes · Encyclopaedia Britannica: black hole
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Black holes suck in everything around them. **Documented:** From a distance, their gravity is like any object of equal mass; only close approaches fall in.

Make it yours

3 swaps so it’s your video

1. Open on a vacuum cleaner icon with a red cross.
2. Add that the Sun is not massive enough to become a black hole.
3. Ask viewers what black hole myth they heard first.

Ready to post

TITLE Black Holes Aren't Cosmic Vacuum Cleaners

DESCRIPTION Far away, a black hole pulls like any equal mass.

HASHTAGS #space #blackhole #science

COVER TEXT Not a vacuum cleaner

PIN Which black hole myth should I bust next?

A Light-Year Is a Distance, Not a Time

BEAT TIMELINE (SECONDS) · TOPIC: LIGHT-YEAR

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	A light-year sounds like time, but it's distance.	Animated icons: clock icon morphing into a ruler	▸ "DISTANCE, NOT TIME"	Firm
3–10	It's the distance light covers in one year, about 9.5 trillion kilometers. That's a lot of zeros.	Big number card: 9,500,000,000,000 km	Highlight "9.5 trillion km"	Clear
10–18	Light could circle Earth about seven times in one second, so a full year of that adds up fast.	Animated diagram: light ray looping around Earth	‖ Pause after "one second"	Amazed
18–27	Astronomers use it because kilometers get silly. The nearest star beyond the Sun is about 4.2 light-years away.	Scale diagram: Sun and nearest star, distance arrow	▸ New card per sentence	Light
27–36	It also means you see that star as it was about 4.2 years ago, because its light needed that long to reach you.	Simple diagram: star-to-eye light path with a calendar	Highlight "4.2 years ago"	Explaining
36–45 CTA	So when someone says light-years to mean a long time, will you correct them? Tell me below.	Ruler icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: What Is a Light-Year? · Encyclopaedia Britannica: light-year
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** A light-year is a measure of time. **Documented:** It's a distance: how far light travels in one year, about 9.5 trillion kilometers.

Make it yours

3 swaps so it's your video

1. Open on a clock that turns into a ruler.
2. Swap the nearest-star example for the Andromeda Galaxy.
3. Ask viewers to share where they heard it used as time.

Ready to post

TITLE A Light-Year Is a Distance, Not a Time

DESCRIPTION Light-years measure how far, not how long.

HASHTAGS #space #lightyear #science

COVER TEXT Light-year = distance

PIN Want a script on parsecs next?

The Mars Rover That Sang Happy Birthday

BEAT TIMELINE (SECONDS) · TOPIC: CURIOSITY ROVER

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	A Mars rover once sang Happy Birthday to itself.	Labelled artist's impression of a rover on red ground	▸ "HAPPY BIRTHDAY?"	Playful
3–10	In 2013, on the first anniversary of its landing, Curiosity hummed the tune by vibrating its sample instrument.	Animated icons: rover with music notes	Highlight "2013"	Light
10–18	It's a car-sized rover that landed in Gale Crater in 2012, lowered on cables from a rocket-powered sky crane.	Simple diagram: sky crane lowering rover on cables	‖ Pause after "Gale Crater"	Clear
18–27	Its main job is reading the rocks, which show that Gale Crater once held lakes of liquid water.	Public-domain agency image of layered Mars rock, credit shown	▸ New card per clause	Explaining
27–36	As of the mid-2020s, Curiosity is **still climbing Mount Sharp, more than ten years after landing**.	Simple diagram: mountain profile with a rover dot	Highlight "still climbing"	Proud
36–45 CTA	That birthday song was a one-off treat. What song would you play on Mars?	Music note icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Mars Science Laboratory / Curiosity rover mission page · Encyclopaedia Britannica: Curiosity (Mars rover)
SHELF-LIFE	**Re-check numbers.** Rover status changes; the voiceover dates it to the mid-2020s.
# RE-CHECK	**Whether Curiosity is still active, and where it is on Mount Sharp.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a birthday candle icon on a red desert.
2. Swap the lake fact for the rover's sky-crane landing story.
3. Ask viewers what song a rover should play.

Ready to post

TITLE The Mars Rover That Sang Happy Birthday

DESCRIPTION Curiosity hummed Happy Birthday on Mars, then went back to work.

HASHTAGS #space #mars #curiosity

COVER TEXT Happy birthday, from Mars

PIN Want the sky-crane landing story next?

How Astronauts Sleep With No Up or Down

BEAT TIMELINE (SECONDS) · TOPIC: SLEEPING IN ORBIT

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	In space, there's no up or down for sleeping.	Animated icons: bed icon rotating in all directions	▸ "NO UP. NO DOWN."	Curious
3–10	On the space station, crew members zip into sleeping bags tied to the wall of a small private cabin.	Simple diagram: small cabin with a bag on the wall	▸ New card per clause	Clear
10–18	Without the bag, they'd drift around and bump into things. Arms often float up in front of them, like a sleepwalking cartoon.	Animated icons: floating figure with arms drifting up	‖ Pause after "things"	Amused
18–27	Fans matter too. Without air flow, the carbon dioxide you breathe out can collect in a bubble around your head.	Simple diagram: head with a CO2 bubble, fan arrows	Highlight "CO2 bubble"	Serious
27–36	And with about 16 sunrises a day, crews use eye masks and schedules to keep their body clocks on track.	Animated icons: sunrise counter, eye mask	▸ New card per clause	Explaining
36–45 CTA	Could you sleep floating, strapped to a wall? Tell me honestly.	Sleeping bag icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA: Sleeping in Space (International Space Station) · Encyclopaedia Britannica: space station, living conditions
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a sleeping bag icon rotating freely.
2. Swap the CO2 beat for how they keep a body clock.
3. Ask viewers how they sleep best on Earth.

Ready to post

TITLE How Astronauts Sleep With No Up or Down

DESCRIPTION Sleeping bags on walls, fans for air and eye masks for sunrises.

HASHTAGS #space #astronaut #sleep

COVER TEXT Sleeping on the wall

PIN Want the space bathroom script next?

Why the Sky Is Blue (It's Not the Ocean)

BEAT TIMELINE (SECONDS) · TOPIC: WHY THE SKY IS BLUE

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The sky is blue because of tiny air molecules.	Stock clip of a clear blue sky	▸ "WHY BLUE?"	Curious
3–10	Sunlight looks white, but it's really a mix of every color, from red to violet.	Animated icons: white beam splitting into a rainbow	▸ New card per clause	Clear
10–18	When it hits the air, the gases scatter short blue wavelengths far more than red ones, spraying blue across the sky.	Simple diagram: blue arrows bouncing off air molecules	Highlight "scatter"	Explaining
18–27	Violet scatters even more, but sunlight has less of it and our eyes are less sensitive to it, so blue wins.	Simple graphic: violet vs blue bars	‖ Pause after "even more"	Light
27–36	At sunset, light crosses much more air, so most blue gets scattered away and reds and oranges reach you.	Simple diagram: long low light path at sunset	▸ New card per clause	Calm
36–45 CTA	And no, the sky isn't reflecting the ocean. Who told you that one?	Ocean icon crossed out + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Space Place: Why Is the Sky Blue? · Encyclopaedia Britannica: Rayleigh scattering
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** The sky is blue because it reflects the ocean. **Documented:** Air molecules scatter blue sunlight more than red; the sky is blue over deserts too.

Make it yours

3 swaps so it's your video

1. Open on a glass of water with a flashlight and milk drops.
2. Add why the sky on Mars looks different.
3. Ask viewers which sky color myth they grew up with.

Ready to post

TITLE Why the Sky Is Blue (It's Not the Ocean)

DESCRIPTION Air scatters blue light most. The ocean isn't the reason.

HASHTAGS #space #science #sky

COVER TEXT Why is the sky blue?

PIN Should I explain red sunsets in detail next?

Saturn Is Less Dense Than Water

BEAT TIMELINE (SECONDS) · TOPIC: SATURN'S DENSITY

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Saturn is less dense than water.	Labelled artist's impression of Saturn	▸ "LESS DENSE THAN WATER"	Curious
3–10	It's about 95 times Earth's mass, but it's so big that its average density is lower than water's.	Scale diagram: Earth and Saturn, mass label	Highlight "95 times"	Clear
10–18	That's why people say Saturn would float in a giant bathtub. It's a fun picture, but it's not the full story.	Animated icons: Saturn bobbing in a cartoon bathtub	‖ Pause after "bathtub"	Playful
18–27	Saturn is mostly hydrogen and helium with no solid surface, and its core is far denser than its outer layers.	Simple diagram: Saturn cut-away with layers labelled	▸ New card per clause	Explaining
27–36	So the float line is a density comparison, not something that could actually happen with a real bathtub.	Simple graphic: density scale with water marked	Highlight "density comparison"	Calm
36–45 CTA	Its rings are even wilder. Want a script on those next?	Saturn icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Saturn Facts · Encyclopaedia Britannica: Saturn (planet)
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Saturn would really float in a big enough bathtub. **Documented:** Its average density is below water's, but it's a gas giant with a dense core; it's only a comparison.

Make it yours

3 swaps so it's your video

1. Open on a rubber duck icon next to Saturn.
2. Compare with Earth, the densest planet in the solar system.
3. Ask viewers what everyday object they'd compare Saturn to.

Ready to post

TITLE Saturn Is Less Dense Than Water

DESCRIPTION Saturn's average density is lower than water's. Here's what that means.

HASHTAGS #space #saturn #planets

COVER TEXT Saturn would float?

PIN Want a Saturn's rings script next?

Earth's Shadow Doesn't Cause Moon Phases

BEAT TIMELINE (SECONDS) · TOPIC: LUNAR PHASES

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Moon phases are not caused by Earth's shadow.	Public-domain agency image of a crescent Moon, credit shown	▸ "NOT THE SHADOW"	Firm
3–10	The Moon is always half lit by the Sun, like Earth's day side and night side.	Simple diagram: Sun, Moon with a lit half	▸ New card per clause	Clear
10–18	As it orbits us over about a month, we see different amounts of that sunlit half from our angle.	Animated diagram: Moon orbit with phase views	Highlight "our angle"	Explaining
18–27	When it sits roughly between us and the Sun, the lit side faces away and we get a new Moon.	Simple diagram: new Moon position, lit side away	‖ Pause after "faces away"	Calm
27–36	Earth's shadow only falls on the Moon during a lunar eclipse, and that's a different event that can turn it reddish.	Simple diagram: Earth's shadow cone reaching the Moon	▸ New card per clause	Clear
36–45 CTA	Which phase is your favorite: thin crescent or full Moon? Comment below.	Crescent and full Moon icons + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Moon Phases · Encyclopaedia Britannica: phase (astronomy)
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Moon phases happen when Earth's shadow covers part of the Moon. **Documented:** Phases come from our changing view of its sunlit half; Earth's shadow only matters in lunar eclipses.

Make it yours

3 swaps so it's your video

1. Demo it with a lamp, a ball and your own head.
2. Swap the eclipse beat for why the crescent's bright side faces the Sun.
3. Ask viewers to name tonight's phase.

Ready to post

TITLE Earth's Shadow Doesn't Cause Moon Phases

DESCRIPTION The Moon is always half lit. Phases are about our angle.

HASHTAGS #space #moon #astronomy

COVER TEXT It's not Earth's shadow

PIN Want the lunar eclipse version next?

Sunlight's Energy May Be Older Than Civilization

BEAT TIMELINE (SECONDS) · TOPIC: ENERGY ESCAPING THE SUN

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Sunlight's energy may be older than human civilization.	Labelled artist's impression of the Sun's layers	▸ "OLDER THAN CIVILIZATION?"	Curious
3–10	Energy is made in the Sun's core, where nuclear fusion turns hydrogen into helium.	Simple diagram: Sun cut-away, core labelled	Highlight "core"	Clear
10–18	But the inside of the Sun is so packed that light can't fly straight out. It gets absorbed and sent off again.	Animated icons: light arrow blocked by crowded dots	▸ New card per sentence	Explaining
18–27	Each time, it heads in a random direction, so the energy zigzags outward in an incredibly slow random walk.	Animated diagram: zigzag path from core outward	Highlight "random walk"	Building
27–36	Estimates vary a lot, but the trip from core to surface may take tens of thousands of years or more.	Big number card with a range bar	‖ Pause after "vary a lot"	Careful
36–45 CTA	Then it crosses to Earth in about eight minutes. Does that change how sunlight feels?	Sun icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Sun Facts, layers of the Sun · Encyclopaedia Britannica: Sun, radiative zone
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a ball bouncing randomly in a crowded box.
2. Compare the timescale with the age of the pyramids.
3. Ask viewers whether the zigzag surprised them.

Ready to post

TITLE Sunlight's Energy May Be Older Than Civilization

DESCRIPTION Energy zigzags out of the Sun for ages before an 8-minute trip.

HASHTAGS #space #sun #physics

COVER TEXT Ancient sunlight energy

PIN Want a script on how fusion works?

The First Image of a Black Hole

BEAT TIMELINE (SECONDS) · TOPIC: M87 BLACK HOLE IMAGE

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	In 2019, the first black hole image was released.	Labelled artist's impression of a glowing ring around a shadow	▸ "2019: FIRST IMAGE"	Dramatic
3–10	It showed the giant black hole at the center of galaxy M87, about 55 million light-years away.	Scale diagram: Milky Way to M87 distance arrow	Highlight "55 million light-years"	Clear
10–18	The black hole itself gives off no light. What you see is a glowing ring of hot gas around its shadow.	Simple diagram: ring and shadow labelled	‖ Pause after "no light"	Explaining
18–27	Getting it took a network of radio telescopes around the world, working together like one Earth-sized dish.	Animated icons: dishes on a globe linked by lines	▸ New card per clause	Building
27–36	That black hole is billions of times the Sun's mass. Its shadow is wider than the orbits of all our planets.	Scale diagram: solar system orbits inside the shadow	Highlight "billions of times"	Amazed
36–45 CTA	Did you know that orange ring was built from real telescope data? Tell me below.	Ring icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Black Hole Image Makes History (M87) · Encyclopaedia Britannica: Event Horizon Telescope
SHELF-LIFE	Evergreen. A historical result; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a blurry ring that slowly sharpens.
2. Compare the shadow's size with the orbits of our planets.
3. Ask viewers what they expected a black hole to look like.

Ready to post

TITLE The First Image of a Black Hole

DESCRIPTION A glowing ring around a shadow, 55 million light-years away.

HASHTAGS #space #blackhole #astronomy

COVER TEXT A real black hole image

PIN Want the Milky Way black hole image next?

Earth Is Speeding Around the Galaxy

BEAT TIMELINE (SECONDS) · TOPIC: GALACTIC YEAR

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	You're moving through the galaxy right now, very fast.	Animated icons: spiral galaxy with a moving Sun dot	▸ "YOU'RE MOVING"	Curious
3–10	The Sun and its planets orbit the center of the Milky Way at roughly 800,000 kilometers per hour.	Simple diagram: Sun's path around the galactic center	Highlight "800,000 km/h"	Clear
10–18	Even at that speed, one full lap takes about 230 million years. Astronomers call it a galactic year.	Big number card: 230,000,000 years	‖ Pause after "one full lap"	Amazed
18–27	One galactic year ago, the first dinosaurs were only just appearing on Earth. Humans have existed for a tiny fraction of one lap.	Simple timeline: dinosaurs at the start, humans at the end	▸ New card per sentence	Building
27–36	By that measure, the Sun is only about 20 galactic years old. It has circled the galaxy around 20 times.	Animated icons: lap counter reaching 20	Highlight "20 galactic years"	Light
36–45 CTA	How many galactic years old are you? Almost zero. Does that make you feel young?	Galaxy icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Milky Way Galaxy · Encyclopaedia Britannica: Milky Way Galaxy, galactic rotation
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a galaxy spiral with a tiny Sun dot moving.
2. Swap dinosaurs for when the first mammals appeared.
3. Ask viewers to guess one lap's length first.

Ready to post

TITLE Earth Is Speeding Around the Galaxy

DESCRIPTION One lap around the Milky Way takes about 230 million years.

HASHTAGS #space #milkyway #galaxy

COVER TEXT One lap: 230 million years

PIN Want a script on where we sit in the galaxy?

Hubble Launched Blurry and Got Glasses

BEAT TIMELINE (SECONDS) · TOPIC: HUBBLE'S MIRROR FLAW

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The Hubble telescope launched with a blurry mirror.	Labelled artist's impression of a space telescope	▸ "BLURRY HUBBLE"	Curious
3–10	It launched in 1990, and its first images came back fuzzy because the mirror was ground slightly wrong.	Simple graphic: fuzzy star next to a sharp star	Highlight "1990"	Clear
10–18	The error was about one fiftieth of a human hair's width, yet that was enough to blur everything.	Scale diagram: hair width with a tiny slice marked	‖ Pause after "hair's width"	Amazed
18–27	In 1993, astronauts flew a servicing mission and installed corrective optics, like giving Hubble glasses.	Animated icons: glasses added to a telescope icon	▸ New card per clause	Light
27–36	After that, the images became sharp. As of the mid-2020s, **Hubble is still observing, more than 30 years after launch**.	Public-domain agency image of a nebula, credit shown	Highlight "still observing"	Proud
36–45 CTA	Which Hubble image is your favorite? Drop it in the comments.	Telescope icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Hubble Space Telescope history · Encyclopaedia Britannica: Hubble Space Telescope
SHELF-LIFE	Re-check numbers. Telescope status changes; the voiceover dates it to the mid-2020s.
# RE-CHECK	**Whether Hubble is still operating, and its years in orbit.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a blurry photo that snaps into focus.
2. Compare the mirror error with a sheet of paper's thickness.
3. Ask viewers which telescope they'd like a script on next.

Ready to post

TITLE Hubble Launched Blurry and Got Glasses

DESCRIPTION A tiny mirror error, fixed in orbit by a servicing mission.

HASHTAGS #space #hubble #telescope

COVER TEXT Hubble needed glasses

PIN Which Hubble image should I explain next?

A Spacesuit Is a Tiny Spaceship You Wear

BEAT TIMELINE (SECONDS) · TOPIC: SPACESUITS

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	A spacesuit is really a tiny spaceship you wear.	Simple graphic: suit outline with a spaceship label	▸ "A WEARABLE SPACESHIP"	Curious
3–10	Outside the station, it has to supply oxygen, hold pressure and handle temperatures from scorching to freezing.	Animated icons: oxygen, pressure gauge, thermometer	▸ New card per job	Clear
10–18	It's white because white reflects sunlight, which helps keep the astronaut from overheating in direct sun.	Simple diagram: sunlight bouncing off a white suit	Highlight "reflects sunlight"	Explaining
18–27	Inside, a cooling garment carries water through thin tubes against the skin, like wearing your own air conditioner.	Simple diagram: body outline with thin tube lines	‖ Pause after "thin tubes"	Light
27–36	Before a spacewalk, astronauts often breathe pure oxygen for a while to clear nitrogen from the blood and avoid the bends.	Animated icons: oxygen mask, bubbles fading	▸ New card per clause	Serious
36–45 CTA	Would you trust a suit with your life for a seven-hour spacewalk? Comment below.	Helmet icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA: Spacesuits and spacewalks overview · Encyclopaedia Britannica: space suit
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a suit icon assembling layer by layer.
2. Swap the cooling beat for how gloves make hands tired.
3. Ask viewers which suit job sounds hardest.

Ready to post

TITLE A Spacesuit Is a Tiny Spaceship You Wear

DESCRIPTION Oxygen, pressure, cooling and sun protection, all in one suit.

HASHTAGS #space #spacesuit #astronaut

COVER TEXT A spaceship you wear

PIN Want a script on spacewalk gloves next?

Solar Eclipses Rely on a Cosmic Coincidence

BEAT TIMELINE (SECONDS) · TOPIC: TOTAL SOLAR ECLIPSES

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	A total solar eclipse depends on a cosmic coincidence.	Labelled artist's impression of a total eclipse	▸ "A COSMIC COINCIDENCE"	Curious
3–10	The Sun is about 400 times wider than the Moon, and also about 400 times farther away.	Scale diagram: Sun, Moon, Earth with two 400x labels	Highlight "400 times"	Clear
10–18	So from Earth they look almost the same size, and the Moon can cover the Sun's bright disk almost perfectly.	Animated icons: Moon disk sliding over the Sun disk	‖ Pause after "same size"	Wonder
18–27	During totality, the sky darkens and the Sun's faint outer atmosphere, the corona, appears as a pale glow.	Simple graphic: black disk with a pale glowing halo	▸ New card per clause	Calm
27–36	Eclipses don't send out harmful rays. The danger is staring at the bright Sun, so use proper eclipse glasses except during totality.	Animated icons: eclipse glasses with a tick	Highlight "eclipse glasses"	Serious
36–45 CTA	Have you ever seen a total eclipse? Where were you?	Eclipse icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Eclipses, solar eclipse safety · Encyclopaedia Britannica: eclipse
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Eclipses give off special harmful rays. **Documented:** The sunlight is the same; the danger is looking at the uncovered Sun without proper eye protection.

Make it yours

3 swaps so it's your video

1. Open with a coin covering a lamp at the right distance.
2. Add why some eclipses leave a ring of fire.
3. Ask viewers where they watched their last eclipse.

Ready to post

TITLE Solar Eclipses Rely on a Cosmic Coincidence

DESCRIPTION The Sun is about 400 times wider, and 400 times farther.

HASHTAGS #space #eclipse #astronomy

COVER TEXT The 400x coincidence

PIN Want a ring-of-fire eclipse script next?

Sunsets on Mars Are Blue

BEAT TIMELINE (SECONDS) · TOPIC: BLUE MARTIAN SUNSETS

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	On Mars, sunsets glow blue.	Public-domain agency image of a Mars sunset, credit shown	▸ "BLUE SUNSETS"	Wonder
3–10	By day, the Martian sky often looks butterscotch or dusty pink, because of fine dust in thin air.	Simple graphic: Mars sky color swatches	▸ New card per clause	Clear
10–18	But at sunset, the area around the Sun turns blue. Rovers have photographed it many times.	Animated icons: rover camera aimed at a blue glow	Highlight "turns blue"	Surprised
18–27	Mars dust scatters blue light forward, close to the direction the light is traveling, so blue gathers around the setting Sun.	Simple diagram: dust grains bending blue light forward	‖ Pause after "forward"	Explaining
27–36	On Earth it's roughly the opposite: blue sky by day, and red and orange near the Sun at sunset.	Split screen: Earth sky and Mars sky	▸ New card per clause	Calm
36–45 CTA	Which sunset would you rather watch, Earth's or Mars's? Tell me below.	Sunset icons + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Mars sunset images from rovers · Encyclopaedia Britannica: Mars (planet), atmosphere
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on an Earth sunset that fades into a blue one.
2. Compare Martian dust grain size with flour.
3. Ask viewers which sunset they'd rather photograph.

Ready to post

TITLE Sunsets on Mars Are Blue

DESCRIPTION Fine Martian dust turns the sky near the setting Sun blue.

HASHTAGS #space #mars #sunset

COVER TEXT Blue sunsets on Mars

PIN Want a script on Mars dust storms next?

The Giant Rising Moon Is an Illusion

BEAT TIMELINE (SECONDS) · TOPIC: MOON ILLUSION

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	A rising Moon isn't actually bigger.	Stock clip of a large Moon rising over buildings	▸ "NOT BIGGER"	Curious
3–10	It looks huge near the horizon, but photos show it's about the same size as when it's high overhead.	Side-by-side photos: low Moon and high Moon, same size	▸ New card per clause	Clear
10–18	If anything, it's slightly farther away at the horizon, because you're looking across part of Earth's width.	Simple diagram: viewer on Earth, two distances to the Moon	Highlight "slightly farther"	Surprised
18–27	This is the Moon illusion, and scientists still debate exactly why our brains fall for it.	Animated icons: brain with a question mark	‖ Pause after "Moon illusion"	Explaining
27–36	One idea: trees and buildings near the horizon make the Moon seem far and big. Up high, there's nothing to compare it with.	Simple graphic: Moon beside trees vs Moon alone	▸ New card per sentence	Calm
36–45 CTA	Try it: hold up your thumb next to the rising Moon tonight. Did it shrink?	Thumb icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: The Moon Illusion · Encyclopaedia Britannica: Moon illusion
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** The Moon is closer when it's near the horizon. **Documented:** It's about the same size, slightly farther; the bigger look is an illusion in our brains.

Make it yours

3 swaps so it's your video

1. Show two photos of the Moon, low and high, side by side.
2. Swap the trees idea for the bent-over view test.
3. Ask viewers to try the thumb test and report back.

Ready to post

TITLE The Giant Rising Moon Is an Illusion

DESCRIPTION The huge horizon Moon is a trick of your brain.

HASHTAGS #space #moon #illusion

COVER TEXT The Moon isn't bigger

PIN Did the thumb test work for you?

Our Nearest Star Neighbor Is Too Faint to See

BEAT TIMELINE (SECONDS) · TOPIC: PROXIMA CENTAURI

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Our nearest star neighbor is too faint to see.	Stock clip of a star field with an empty circle	▸ "NEAREST, BUT INVISIBLE"	Curious
3–10	It's Proxima Centauri, about 4.2 light-years away, a small red dwarf much dimmer than the Sun.	Scale diagram: Sun beside a small red dwarf	Highlight "4.2 light-years"	Clear
10–18	Its brighter neighbors, Alpha Centauri A and B, look like one bright star from the Southern Hemisphere.	Simple sky chart: one bright point, labelled A and B	‖ Pause after "neighbors"	Explaining
18–27	Red dwarfs are thought to be the most common stars in our galaxy, and nearly all are too dim to see.	Animated icons: many red dots, few bright dots	▸ New card per clause	Building
27–36	As of the mid-2020s, **at least one planet is confirmed orbiting Proxima**, in the zone where liquid water could exist.	Labelled artist's impression: rocky planet near a red star	Highlight "at least one planet"	Wonder
36–45 CTA	If we could visit one star system, should it be this one? Comment below.	Red star icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Proxima Centauri b · Encyclopaedia Britannica: Proxima Centauri
SHELF-LIFE	**Re-check numbers.** Planet counts change as discoveries are confirmed; the voiceover hedges it.
# RE-CHECK	**Number of confirmed planets orbiting Proxima Centauri.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a sky chart with an empty circle where Proxima is.
2. Compare its size with Jupiter instead of the Sun.
3. Ask viewers whether they'd send a probe there.

Ready to post

TITLE Our Nearest Star Neighbor Is Too Faint to See

DESCRIPTION The closest star beyond the Sun is a dim red dwarf.

HASHTAGS #space #stars #proximacentauri

COVER TEXT The closest star you can't see

PIN Want a script on Alpha Centauri A and B?

Magnetars: Dead Stars That Are Monster Magnets

BEAT TIMELINE (SECONDS) · TOPIC: MAGNETARS

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Some dead stars are monster magnets.	Labelled artist's impression of a magnetar with field lines	▸ "MONSTER MAGNETS"	Dramatic
3–10	They're called magnetars, a rare kind of neutron star, the crushed core left after a giant star explodes.	Simple diagram: giant star, explosion, small core	▸ New card per clause	Clear
10–18	Their magnetic fields are the strongest we know of, up to a thousand trillion times Earth's.	Big number card: 1,000,000,000,000,000 x	Highlight "strongest we know of"	Amazed
18–27	Scientists have calculated that near one, the field would distort the atoms in your body. You'd never get close enough to notice.	Animated icons: atom stretched into an oval	‖ Pause after "your body"	Serious
27–36	Sometimes their crust cracks in a starquake, releasing a flash of energy that instruments on Earth have detected from across the galaxy.	Animated icons: cracking crust, flash reaching Earth	▸ New card per clause	Building
36–45 CTA	Which is scarier to you, a black hole or a magnetar? Tell me below.	Magnet and black hole icons + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Magnetars (neutron stars) · Encyclopaedia Britannica: magnetar
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a fridge magnet next to a glowing star icon.
2. Swap the starquake for how fast some neutron stars spin.
3. Ask viewers to rank the scariest objects in space.

Ready to post

TITLE Magnetars: Dead Stars That Are Monster Magnets

DESCRIPTION Magnetars have the strongest magnetic fields we know of.

HASHTAGS #space #magnetar #physics

COVER TEXT The universe's monster magnets

PIN Want the pulsar script next?

We Can See Farther Than 13.8 Billion Light-Years

BEAT TIMELINE (SECONDS) · TOPIC: OBSERVABLE UNIVERSE

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	We can see farther than 13.8 billion light-years.	Animated icons: zooming star field	▸ "FARTHER THAN 13.8B"	Curious
3–10	The universe is about 13.8 billion years old, so you'd guess that's as far as we can see.	Big number card: 13.8 billion years	▸ New card per clause	Clear
10–18	But space itself has been stretching the whole time, carrying distant galaxies farther away while their light traveled toward us.	Animated diagram: dots on an inflating balloon	Highlight "stretching"	Explaining
18–27	So the most distant regions we can see are now estimated to be about 46 billion light-years away.	Scale diagram: Earth at center, 46 billion ly radius	‖ Pause after "estimated"	Amazed
27–36	That makes the observable universe about 93 billion light-years across. Beyond that, there's likely more universe we simply can't see.	Simple graphic: circle labelled observable, faded area beyond	▸ New card per sentence	Wonder
36–45 CTA	Does an edge we can't see beyond excite you or scare you? Tell me below.	Circle icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: The Big Bang and the expanding universe · Encyclopaedia Britannica: observable universe
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** We can only see 13.8 billion light-years in any direction. **Documented:** Space expanded while light traveled; the observable universe is about 93 billion light-years across.

Make it yours

3 swaps so it's your video

1. Open on a balloon with dots drawn on it, inflating.
2. Swap the edge question for where the oldest light comes from.
3. Ask viewers what they think lies beyond.

Ready to post

TITLE We Can See Farther Than 13.8 Billion Light-Years

DESCRIPTION Expanding space stretches the observable universe to about 93 billion light-years.

HASHTAGS #space #universe #cosmology

COVER TEXT Bigger than its age?

PIN Want a script on the oldest light we can see?

New Horizons Found a Giant Heart on Pluto

BEAT TIMELINE (SECONDS) · TOPIC: NEW HORIZONS AT PLUTO

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Pluto has a giant heart on its surface.	Public-domain agency image of Pluto's heart region, credit shown	▸ "PLUTO HAS A HEART"	Warm
3–10	We saw it in 2015, when New Horizons flew past Pluto after about nine years in space.	Animated icons: generic probe on a dashed path to Pluto	Highlight "2015"	Clear
10–18	Before that, Pluto was just a few blurry pixels. Suddenly we had mountains, plains and a thin hazy atmosphere.	Simple graphic: pixel blob vs detailed sphere	▸ New card per sentence	Amazed
18–27	The heart's left side is a huge plain of frozen nitrogen, slowly churning like a lava lamp in slow motion.	Animated icons: slow bubbling cells	‖ Pause after "frozen nitrogen"	Wonder
27–36	Some of its mountains are made of water ice, which is rock hard at Pluto's temperatures, far below freezing.	Simple diagram: ice mountain with a thermometer	Highlight "water ice"	Explaining
36–45 CTA	Should Pluto be called a planet again? Make your case below.	Pluto icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: New Horizons mission page · Encyclopaedia Britannica: New Horizons, and Pluto
SHELF-LIFE	Evergreen. A historical flyby; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on the old blurry Pluto pixels, then the sharp view.
2. Swap the ice mountains for Pluto's blue hazy sky.
3. Ask viewers to vote on Pluto's planet status.

Ready to post

TITLE New Horizons Found a Giant Heart on Pluto

DESCRIPTION A 2015 flyby turned Pluto from blurry pixels into a real world.

HASHTAGS #space #pluto #newhorizons

COVER TEXT Pluto has a heart

PIN Should I cover Pluto's moon Charon next?

Why Astronauts Work Out Two Hours a Day

BEAT TIMELINE (SECONDS) · TOPIC: BONE AND MUSCLE LOSS IN SPACE

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Space can weaken an astronaut's bones.	Simple diagram: bone outline with a falling arrow	▸ "WEAKER BONES"	Serious
3–10	Without gravity pulling on them, bones carry less weight, and the body starts breaking some bone down.	Animated icons: bone icon losing small pieces	▸ New card per clause	Clear
10–18	Astronauts can lose roughly one percent of bone mass in some areas each month, especially the hips and spine.	Simple diagram: skeleton with hips and spine marked	Highlight "one percent"	Explaining
18–27	Muscles shrink too, because floating takes so little effort. Even the heart doesn't have to work as hard.	Animated icons: muscle shrinking, heart icon	‖ Pause after "shrink too"	Calm
27–36	So crews exercise about two hours a day on a treadmill, a bike and a machine that mimics lifting weights.	Animated icons: treadmill, bike, weight machine	▸ New card per item	Building
36–45 CTA	Could you work out two hours every day for six months? Be honest.	Dumbbell icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA: Human Research Program, bone and muscle loss · Encyclopaedia Britannica: weightlessness, physiological effects
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a dumbbell icon floating away.
2. Swap the heart beat for balance problems after landing.
3. Ask viewers what workout they would pick in orbit.

Ready to post

TITLE Why Astronauts Work Out Two Hours a Day

DESCRIPTION Without gravity, bones and muscles weaken unless you train daily.

HASHTAGS #space #astronaut #fitness

COVER TEXT Two hours, every day

PIN Want a script on landing back on Earth?

The North Star Isn't the Brightest Star

BEAT TIMELINE (SECONDS) · TOPIC: POLARIS

0 3 10 18 27 36 45

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The North Star is not the brightest star.	Stock clip of a night sky, one star circled	▸ "NOT THE BRIGHTEST"	Firm
3–10	Polaris is only around the fiftieth brightest star in the night sky. Sirius is far brighter.	Simple graphic: brightness ranking bars	Highlight "around the fiftieth"	Clear
10–18	Polaris matters because it sits almost directly above Earth's North Pole, so it barely moves while other stars circle around it.	Animated diagram: star trails circling one fixed point	‖ Pause after "North Pole"	Explaining
18–27	That made it a trusted guide for travelers. Find it, and you've found north, and your latitude too.	Simple diagram: Polaris height angle equals latitude	▸ New card per clause	Building
27–36	It won't be the pole star forever. Earth's axis slowly wobbles, so over thousands of years other stars take the job.	Animated diagram: wobbling axis tracing a circle	Highlight "slowly wobbles"	Calm
36–45 CTA	Can you find Polaris from where you live? Tell me below.	Star icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Space Place: North Star · Encyclopaedia Britannica: Polaris
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** The North Star is the brightest star in the sky. **Documented:** Polaris is only moderately bright; it's useful because it sits nearly above the North Pole.

Make it yours

3 swaps so it's your video

1. Show how to find Polaris from the Big Dipper.
2. Swap the wobble beat for how sailors used it at sea.
3. Ask viewers from the south what they navigate by.

Ready to post

TITLE The North Star Isn't the Brightest Star

DESCRIPTION Polaris isn't the brightest. It's the one that stays put.

HASHTAGS #space #northstar #stargazing

COVER TEXT Not the brightest star

PIN Want a script on finding south at night?

Jupiter’s Great Red Spot Is Shrinking

BEAT TIMELINE (SECONDS) · TOPIC: GREAT RED SPOT

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Jupiter's Great Red Spot is shrinking.	Labelled artist's impression of Jupiter's red storm	▸ “IT'S SHRINKING”	Curious
3–10	It's a giant storm that has been raging for at least 150 years, and maybe much longer.	Simple timeline: 150-year bar with a question mark	Highlight “150 years”	Clear
10–18	In the late 1800s, observers described it as wide enough to hold about three Earths side by side.	Scale diagram: three Earths inside a large oval	▸ New card per clause	Building
18–27	As of the mid-2020s, it's only **about 1.3 times Earth's width**, and it has become rounder.	Scale diagram: smaller oval with one Earth inside	Highlight “1.3 times”	Surprised
27–36	Its winds still blow at hundreds of kilometers per hour, much faster than the strongest hurricanes on Earth.	Animated icons: swirling arrows with a speed label	‖ Pause after “still blow”	Dramatic
36–45	Nobody knows exactly why it's shrinking. Scientists are still studying how it feeds on smaller storms.	Animated icons: small storms merging into a big one	▸ New card per sentence	Explaining
45–54	So any size you quote for it needs a date attached. Check before you post, every time.	Calendar icon next to a ruler icon	Hold card 1 s	Wry
54–60 CTA	Do you think the Great Red Spot will ever vanish? Comment below.	Storm icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Jupiter's Great Red Spot · Encyclopaedia Britannica: Great Red Spot
SHELF-LIFE	**Re-check numbers.** The storm's size keeps changing; the voiceover dates the width to the mid-2020s.
# RE-CHECK	**The Great Red Spot's current width compared with Earth.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it’s your video

1. Open on Earth icons lined up inside the storm, then one fading out.
2. Swap the winds beat for how the storm's color may come from sunlight chemistry.
3. Ask viewers to guess the width before the reveal.

Ready to post

TITLE Jupiter's Great Red Spot Is Shrinking

DESCRIPTION The giant storm was once about three Earths wide.

HASHTAGS #space #jupiter #greatredspot

COVER TEXT Jupiter's storm is shrinking

PIN Want a script on Jupiter's polar cyclones?

What Really Happens to a Body in Space

BEAT TIMELINE (SECONDS) · TOPIC: VACUUM EXPOSURE

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Space wouldn't make you explode.	Stock clip of a starfield with a suit icon	▸ "NO EXPLODING"	Firm
3–10	Without a suit in vacuum, you wouldn't burst or freeze solid in an instant.	Animated icons: burst and ice icons crossed out	▸ New card per clause	Clear
10–18	The real danger is oxygen. With no air pressure, your lungs empty and your blood quickly runs low on oxygen.	Simple diagram: lungs with outward arrows, O2 meter falling	Highlight "oxygen"	Serious
18–27	Studies of accidents in vacuum chambers suggest you'd stay conscious for only about 10 to 15 seconds.	Big number card: 10–15 seconds	‖ Pause after "conscious"	Serious
27–36	Your skin and tissues would swell, but your skin is strong enough to hold you together. Your blood stays inside you.	Simple diagram: body outline swelling slightly	▸ New card per sentence	Explaining
36–45	Saliva on your tongue could start to bubble, because liquids boil at very low pressure even when they aren't hot.	Animated icons: water drop with bubbles, pressure gauge at zero	Highlight "very low pressure"	Curious
45–54	Animal studies and rare accidents suggest a quick rescue, within about a minute, may be survivable.	Animated icons: clock at 60 s, rescue arrow	‖ Pause after "quick rescue"	Calm
54–60 CTA	Which space myth did movies teach you? Comment below.	Myth icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA: Ask an Astrophysicist, human body in a vacuum · Encyclopaedia Britannica: space medicine, decompression
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** In space without a suit, you'd explode or your blood would boil instantly. **Documented:** You'd swell and pass out in about 15 seconds; skin holds you together.

Make it yours

3 swaps so it's your video

1. Open on a countdown from 15 over a starfield.
2. Swap the saliva beat for why holding your breath is dangerous.
3. Ask viewers which space myth they believed longest.

Ready to post

TITLE What Really Happens to a Body in Space

DESCRIPTION No exploding, no instant freezing. The real danger is oxygen.

HASHTAGS #space #science #astronaut

COVER TEXT Would you explode in space?

PIN Which space myth should I check next?

The Sun Has an Eleven-Year Cycle

BEAT TIMELINE (SECONDS) · TOPIC: SOLAR CYCLE

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The Sun has an eleven-year mood cycle.	Labelled artist's impression of the Sun with sunspots	▸ "11-YEAR CYCLE"	Curious
3–10	Every eleven years or so, it swings from quiet to stormy and back. It's called the solar cycle.	Simple chart: wave line with peaks about 11 years apart	▸ New card per sentence	Clear
10–18	At its peak, more dark sunspots appear. They're cooler patches where tangled magnetic fields hold back the heat.	Simple diagram: sunspot with tangled field lines	Highlight "cooler patches"	Explaining
18–27	Some sunspots are wider than Earth, yet they only look dark next to the brighter surface around them.	Scale diagram: Earth beside a sunspot	‖ Pause after "wider than Earth"	Amazed
27–36	More sunspots also mean more solar flares and huge clouds of charged particles thrown out into space.	Animated icons: flare burst, particle cloud leaving the Sun	▸ New card per clause	Building
36–45	When those clouds reach Earth, they can trigger bright auroras far from the poles and disturb satellites and radio signals.	Stock clip of an aurora over a landscape	Highlight "auroras"	Wonder
45–54	Then the Sun calms down again, and for a while sunspots can almost disappear. The magnetic field also flips each cycle.	Animated icons: spots fading, magnet icon flipping	‖ Pause after "calms down"	Calm
54–60 CTA	Have you ever seen an aurora? Tell me where.	Aurora icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Solar Cycle · Encyclopaedia Britannica: sunspot, and solar cycle
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a calm Sun icon turning spotty over a counter.
2. Swap the aurora beat for how flares affect GPS signals.
3. Ask viewers if they'd travel to see an aurora.

Ready to post

TITLE The Sun Has an Eleven-Year Cycle

DESCRIPTION Every eleven years or so, the Sun swings from quiet to stormy.

HASHTAGS #space #sun #aurora

COVER TEXT The Sun's 11-year cycle

PIN Want an aurora explainer next?

Saturn's Moon Count Keeps Changing

BEAT TIMELINE (SECONDS) · TOPIC: MOONS OF SATURN

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Saturn has so many moons, the count keeps changing.	Saturn icon beside a fast-spinning number counter	▸ "THE COUNT KEEPS CHANGING"	Curious
3–10	At last count, Saturn has **well over two hundred** confirmed moons. That's more than any other planet we know.	Public-domain agency image of Saturn, credit shown	‖ Pause 0.5 s after "moons"	Confident
10–18	Most of them are tiny. Many are just a few kilometers across, more like captured rocks than round worlds.	Scale diagram: tiny moon next to a city outline	▸ New card per sentence	Light
18–27	Astronomers find them by stacking many telescope images until faint dots moving with Saturn stand out.	Animated icons: stacked frames, one dot circled	Highlight "faint dots"	Explaining
27–36	But a few are giants. Titan is bigger than the planet Mercury and has a thick, hazy atmosphere.	Scale diagram: Titan beside Mercury, both labelled	‖ Pause after "giants"	Building
36–45	And little Enceladus sprays jets of water ice into space from cracks near its south pole.	Artist's impression of Enceladus jets, labelled	▸ "WATER ICE JETS"	Wonder
45–54	So the next time someone quotes a moon count, check the date. The number keeps climbing.	Counter icon next to a calendar icon	Hold card 1 s	Wry
54–60 CTA	Which moon would you visit first, Titan or Enceladus? Tell me below.	Titan and Enceladus icons + comment bubble	Hold last card 1.5 s	Warm, direct

VERIFY WITH	NASA Science: Saturn Moons · Encyclopaedia Britannica: Saturn (planet), moons section
SHELF-LIFE	**Re-check numbers.** The moon count changes as new moons are confirmed; the voiceover hedges it.
# RE-CHECK	**Number of confirmed moons of Saturn (and of Jupiter, if you compare them).**
MYTH CHECK	**Popular version:** Jupiter has the most moons. **Documented:** Saturn moved ahead after batches of new discoveries in the 2020s; both counts can still change.

Make it yours

3 swaps so it's your video

1. Open on a counter spinning upward instead of Saturn itself.
2. Swap Enceladus for Hyperion and its sponge-like, cratered surface.
3. Change the CTA: ask viewers to guess today's count first.

Ready to post

TITLE Saturn's Moon Count Keeps Changing

DESCRIPTION Why Saturn's moon total is a moving target.

HASHTAGS #space #saturn #astronomy

COVER TEXT How many moons? Check again.

PIN Which space number should I re-check next?

The Space Signal Nicknamed Little Green Men

BEAT TIMELINE (SECONDS) · TOPIC: PULSARS

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	In 1967, a strange signal ticked like a clock.	Animated icons: radio dish with pulse marks	▸ "TICK. TICK. TICK."	Mysterious
3–10	A radio telescope picked up pulses arriving about every 1.3 seconds, steady and precise.	Simple chart: evenly spaced pulse spikes	Highlight "1.3 seconds"	Clear
10–18	The team half-jokingly nicknamed it LGM-1, for little green men, because nothing natural seemed that regular.	Simple graphic: LGM-1 label with a wink icon	‖ Pause after "LGM-1"	Playful
18–27	It turned out to be a pulsar: a spinning neutron star, the crushed core of an exploded star.	Labelled artist's impression of a pulsar with two beams	▸ New card per clause	Explaining
27–36	Pulsars shoot beams of radio waves from their magnetic poles. As they spin, the beams sweep past Earth like a lighthouse.	Animated diagram: rotating beams sweeping past Earth	Highlight "like a lighthouse"	Building
36–45	Some spin hundreds of times every second, and their ticks are so regular they can rival atomic clocks.	Animated icons: fast spinning dot beside a clock	‖ Pause after "every second"	Amazed
45–54	Scientists now use networks of pulsars to search for faint ripples in space itself, called gravitational waves.	Simple diagram: several pulsars linked to Earth by lines	▸ New card per clause	Wonder
54–60 CTA	Would you have guessed aliens too? Be honest below.	Pulsar icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Neutron Stars and pulsars · Encyclopaedia Britannica: pulsar
SHELF-LIFE	**Evergreen.** A historical discovery and settled physics; nothing here changes year to year.
MYTH CHECK	**Popular version:** The first pulsar signal was an alien message. **Documented:** The LGM-1 nickname was a joke; pulsars are spinning neutron stars with sweeping radio beams.

Make it yours

3 swaps so it's your video

1. Open on a metronome ticking over a starfield.
2. Swap the gravitational-wave beat for the fastest-spinning pulsars.
3. Ask viewers what they'd have guessed in 1967.

Ready to post

TITLE The Space Signal Nicknamed Little Green Men

DESCRIPTION A ticking radio signal turned out to be a spinning dead star.

HASHTAGS #space #pulsar #astronomy

COVER TEXT Little green men?

PIN Want the magnetar script next?

We've Found Thousands of Planets Beyond the Sun

BEAT TIMELINE (SECONDS) · TOPIC: EXOPLANET COUNT

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	We've found thousands of planets around other stars.	Stock clip of a star field, dots lighting up	▸ "THOUSANDS OF WORLDS"	Wonder
3–10	As of the mid-2020s, the confirmed count is **more than 5,000**, and it keeps climbing.	Counter icon climbing past 5,000	Highlight "more than 5,000"	Clear
10–18	Just over thirty years ago, we didn't know of a single planet around a Sun-like star.	Simple timeline: empty bar, then rising line	‖ Pause after "thirty years ago"	Building
18–27	Most are found by watching a star dim slightly when a planet passes in front of it, like a tiny eclipse.	Simple chart: light curve with a small dip	▸ New card per clause	Explaining
27–36	Others show up as a wobble, as the planet's gravity tugs its star back and forth.	Animated diagram: star wobbling as a planet orbits	Highlight "wobble"	Clear
36–45	Many are nothing like our planets: giant worlds hugging their stars, or planets between Earth and Neptune in size.	Scale diagram: Earth, mid-size planet, Neptune	▸ New card per type	Curious
45–54	Because so many stars seem to have planets, there may be more planets than stars in our galaxy.	Animated icons: stars each gaining small planet dots	‖ Pause after "planets"	Amazed
54–60 CTA	Which kind of alien world would you visit first? Tell me below.	Planet icons + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Exoplanets, discoveries dashboard · Encyclopaedia Britannica: exoplanet
SHELF-LIFE	**Re-check numbers.** The confirmed count rises every year; the voiceover hedges it.
# RE-CHECK	**Current number of confirmed exoplanets.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a star dimming with a tiny dot crossing it.
2. Swap the wobble method for direct imaging.
3. Ask viewers to name a planet type they'd visit.

Ready to post

TITLE We've Found Thousands of Planets Beyond the Sun

DESCRIPTION From zero known to thousands confirmed in about thirty years.

HASHTAGS #space #exoplanets #astronomy

COVER TEXT Thousands of new worlds

PIN Want a script on the strangest exoplanet?

Cassini's Final Dive Into Saturn

BEAT TIMELINE (SECONDS) · TOPIC: CASSINI GRAND FINALE

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	A spacecraft ended its mission by diving into Saturn.	Labelled artist's impression of a probe above Saturn	▸ "THE FINAL DIVE"	Dramatic
3–10	Cassini reached Saturn in 2004 and spent about thirteen years studying the planet, its rings and moons.	Simple timeline: 2004 to 2017	Highlight "thirteen years"	Clear
10–18	It carried the Huygens probe to Titan, and found water jets on Enceladus hinting at a hidden ocean.	Animated icons: probe dropping to a moon, icy jets	▸ New card per clause	Building
18–27	By 2017, it was running out of fuel. Left alone, it might one day have crashed into one of those moons.	Animated icons: fuel gauge near empty	‖ Pause after "fuel"	Serious
27–36	Mission teams didn't want any Earth microbes on board to reach a moon that might support life.	Simple graphic: microbe icon with a stop sign	Highlight "Earth microbes"	Explaining
36–45	So they flew it through the gap between Saturn and its rings 22 times, gathering data no probe had collected before.	Animated diagram: orbit loops threading the ring gap	▸ New card per clause	Amazed
45–54	In September 2017, Cassini plunged into Saturn's atmosphere, sending data until it broke apart.	Animated icons: probe streak entering a planet's clouds	‖ Pause after "plunged"	Solemn
54–60 CTA	Was that a fitting ending for Cassini? Tell me what you think.	Probe icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Cassini mission, Grand Finale · Encyclopaedia Britannica: Cassini-Huygens
SHELF-LIFE	**Evergreen.** A finished mission; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a probe icon falling toward a ringed planet.
2. Swap the Huygens fact for Cassini's ring images.
3. Ask viewers how they'd end a space mission.

Ready to post

TITLE Cassini's Final Dive Into Saturn

DESCRIPTION Why a probe was sent into Saturn on purpose after 13 years.

HASHTAGS #space #cassini #saturn

COVER TEXT A planned final dive

PIN Want the Huygens landing on Titan next?

Space Is Closer Than You Think

BEAT TIMELINE (SECONDS) · TOPIC: PEOPLE WHO HAVE FLOWN TO SPACE

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Space is closer than you think.	Stock clip of clouds from high altitude	▸ "CLOSER THAN YOU THINK"	Curious
3–10	Space is often said to begin about 100 kilometers up, roughly an hour's drive straight up.	Simple diagram: car on a vertical road, 100 km mark	Highlight "100 km"	Clear
10–18	That line is a convention, not a wall. Some agencies use about 80 kilometers instead, so definitions differ.	Simple diagram: two dashed lines at 80 and 100 km	▸ New card per sentence	Explaining
18–27	The hard part isn't height. To stay in orbit, you need to fly sideways at about 28,000 kilometers per hour.	Animated diagram: object circling Earth, speed label	Highlight "28,000 km/h"	Building
27–36	That's why only a small group of people have ever gone. As of the mid-2020s, it's **more than 600 people**.	Counter icon passing 600	‖ Pause after "ever gone"	Calm
36–45	The count depends on which boundary you use, and it grows with every crewed flight.	Simple graphic: two counters side by side	▸ New card per clause	Clear
45–54	Compared with more than eight billion people on Earth, that's a tiny club. Most of them flew to orbit, not just above the line.	Scale diagram: tiny dot cluster beside a huge crowd	Highlight "tiny club"	Wonder
54–60 CTA	If you got a free seat, would you take it? Comment below.	Seat icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	Encyclopaedia Britannica: Kármán line, and astronaut · NASA: Human Spaceflight overview
SHELF-LIFE	**Re-check numbers.** The count grows with every crewed flight; the voiceover hedges it.
# RE-CHECK	**Number of people who have flown to space, and which boundary the count uses.**
MYTH CHECK	**Popular version:** Space starts hundreds of kilometers above Earth. **Documented:** A common boundary is about 100 kilometers; the hard part is orbital speed, not height.

Make it yours

3 swaps so it's your video

1. Open on a car icon driving straight up for one hour.
2. Swap the orbit-speed beat for how long the climb to orbit takes.
3. Ask viewers whether they'd fly suborbital or to orbit.

Ready to post

TITLE Space Is Closer Than You Think

DESCRIPTION Space begins about 100 km up. Staying there is the hard part.

HASHTAGS #space #astronaut #spaceflight

COVER TEXT An hour's drive up

PIN Want a script on how fast orbit really is?

You Can't See the Great Wall From the Moon

BEAT TIMELINE (SECONDS) · TOPIC: GREAT WALL FROM SPACE

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	You can't see the Great Wall from the Moon.	Stock clip of the Great Wall from the ground	▸ "NOT FROM THE MOON"	Firm
3–10	It's one of the most repeated space myths, and it was in print long before anyone flew to space.	Simple graphic: old book icon with a myth stamp	▸ New card per clause	Clear
10–18	The wall is very long, but in most places it's only a few meters wide, about as wide as a road.	Scale diagram: wall width next to a road	Highlight "a few meters wide"	Explaining
18–27	From the Moon, about 384,000 kilometers away, even whole continents look small. A road-width line is far too thin to see.	Public-domain agency image of Earth from far away, credit shown	‖ Pause after "look small"	Calm
27–36	Even from the space station, about 400 kilometers up, astronauts say it's extremely hard to spot without help.	Simple diagram: station orbit 400 km above Earth	▸ New card per clause	Clear
36–45	Its color blends in with the land around it, which makes it harder to pick out than its length suggests.	Simple graphic: grey line fading into brown terrain	Highlight "blends in"	Explaining
45–54	Meanwhile, city lights at night, big airports and highways can stand out clearly from orbit.	Public-domain agency image of city lights at night, credit shown	▸ New card per item	Wonder
54–60 CTA	Where did you first hear the Great Wall myth? Comment below.	Wall icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Earth Observatory: Great Wall from space · Encyclopaedia Britannica: Great Wall of China
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** The Great Wall of China is visible from the Moon. **Documented:** It's far too narrow to see from the Moon and very hard to spot even from low orbit.

Make it yours

3 swaps so it's your video

1. Open on a zoom-out from a road to the whole Earth.
2. Swap the city lights beat for what astronauts say is easy to spot.
3. Ask viewers where they first heard the myth.

Ready to post

TITLE You Can't See the Great Wall From the Moon

DESCRIPTION Too narrow to see from the Moon, hard from orbit.

HASHTAGS #space #mythbusting #earth

COVER TEXT Visible from space? No.

PIN Which space myth should I bust next?

The Closest Planet to the Sun Isn't the Hottest

BEAT TIMELINE (SECONDS) · TOPIC: MERCURY'S TEMPERATURES

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The closest planet to the Sun isn't the hottest.	Simple diagram: Sun, Mercury and Venus in a row	▸ "CLOSEST ≠ HOTTEST"	Curious
3–10	Mercury sits closest, yet Venus is hotter, at around 465 degrees Celsius, day and night.	Animated icons: two thermometers, Venus higher	Highlight "465 °C"	Surprised
10–18	Mercury has almost no atmosphere, so heat escapes easily. Its day side can reach about 430 degrees Celsius.	Labelled artist's impression of Mercury's cratered surface	▸ New card per sentence	Clear
18–27	But its night side drops to about minus 180 degrees, one of the biggest temperature swings in the solar system.	Simple chart: temperature bar from 430 to minus 180	‖ Pause after "minus 180"	Amazed
27–36	Venus is the opposite. Its thick carbon dioxide atmosphere traps heat like a greenhouse with the windows shut.	Simple diagram: heat arrows trapped under a thick layer	Highlight "traps heat"	Explaining
36–45	That's hot enough to melt lead, and it barely cools at night, because the heat has nowhere to go.	Animated icons: lead block melting	▸ New card per clause	Building
45–54	Even stranger, scientists have found evidence of water ice in shadowed craters near Mercury's poles.	Simple diagram: polar crater in shadow with ice label	‖ Pause after "water ice"	Wonder
54–60 CTA	Which would you rather survive for one minute, Mercury or Venus? Tell me below.	Planet icons + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Mercury Facts, and Venus Facts · Encyclopaedia Britannica: Mercury (planet)
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Mercury is the hottest planet because it's closest to the Sun. **Documented:** Venus is hotter; its thick atmosphere traps heat, while Mercury's night side gets extremely cold.

Make it yours

3 swaps so it's your video

1. Open on two thermometers labelled Mercury and Venus.
2. Swap the ice beat for how long a Mercury day lasts.
3. Ask viewers to guess the hottest planet first.

Ready to post

TITLE The Closest Planet to the Sun Isn't the Hottest

DESCRIPTION Mercury bakes and freezes. Venus just bakes, day and night.

HASHTAGS #space #mercury #planets

COVER TEXT Closest isn't hottest

PIN Want a script on Venus's clouds next?

Io: The Moon Covered in Volcanoes

BEAT TIMELINE (SECONDS) · TOPIC: IO'S VOLCANOES

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Jupiter's moon Io is covered in erupting volcanoes.	Public-domain agency image of Io, credit shown	▸ "A VOLCANO MOON"	Dramatic
3–10	It's thought to be the most volcanically active world we know of, with hundreds of active volcanoes.	Simple graphic: Io disk with many volcano dots	Highlight "hundreds"	Clear
10–18	Some lava lakes are larger than cities, and some plumes shoot material hundreds of kilometers above the surface.	Labelled artist's impression of a tall volcanic plume	▸ New card per clause	Amazed
18–27	Io is about the size of our Moon, so where does all that heat come from? Not from the Sun.	Scale diagram: Io beside our Moon	‖ Pause after "come from"	Curious
27–36	Jupiter's huge gravity, plus tugs from neighboring moons, keeps squeezing Io as it moves along its slightly stretched orbit.	Animated diagram: oval orbit, Io squeezed near Jupiter	Highlight "squeezing"	Explaining
36–45	Its solid surface rises and falls by up to about 100 meters, and that flexing heats the inside like a squeezed stress ball.	Animated icons: stress ball squeezed, heat lines	▸ New card per clause	Building
45–54	The result is a yellow, orange and black world painted by sulfur, constantly resurfaced by fresh lava.	Simple graphic: yellow, orange and black color swatches	‖ Pause after "sulfur"	Wonder
54–60 CTA	Would you risk a flyby of Io for the view? Tell me below.	Io icon + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Io · Encyclopaedia Britannica: Io (moon)
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a stress ball being squeezed over Jupiter.
2. Swap the lava lake beat for Io's sulfur snow.
3. Ask viewers which moon they'd least like to land on.

Ready to post

TITLE Io: The Moon Covered in Volcanoes

DESCRIPTION Jupiter's gravity squeezes Io so hard that it's covered in volcanoes.

HASHTAGS #space #io #jupiter

COVER TEXT A moon of volcanoes

PIN Want a script on Jupiter's other big moons?

Why the Sun's Atmosphere Is Hotter Than Its Surface

BEAT TIMELINE (SECONDS) · TOPIC: CORONAL HEATING

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The Sun's outer atmosphere is hotter than its surface.	Labelled artist's impression of the Sun's corona	▸ "HOTTER OUTSIDE?"	Curious
3–10	The visible surface is about 5,500 degrees Celsius. That's already hot enough to melt any metal we know.	Big number card: about 5,500 °C	Highlight "5,500 °C"	Clear
10–18	But the corona, the thin glowing atmosphere above it, reaches more than a million degrees.	Big number card: 1,000,000+ °C	‖ Pause after "corona"	Amazed
18–27	That's like walking away from a campfire and feeling the air get hotter. It shouldn't work that way.	Animated icons: campfire, thermometer rising with distance	▸ New card per sentence	Playful
27–36	This is called the coronal heating problem, and scientists have been working on it for decades.	Simple graphic: puzzle piece over the Sun	Highlight "coronal heating problem"	Explaining
36–45	Leading ideas involve magnetic fields: waves carrying energy upward, and countless small explosions where magnetic lines snap and reconnect.	Simple diagram: magnetic loops snapping with tiny flashes	▸ New card per idea	Building
45–54	As of the mid-2020s, spacecraft flying close to the Sun are **still testing these ideas**.	Animated icons: generic probe near a glowing Sun	‖ Pause after "mid-2020s"	Calm
54–60 CTA	Which space mystery should I cover next? Tell me below.	Question mark icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Sun, the corona · Encyclopaedia Britannica: corona (astronomy)
SHELF-LIFE	**Re-check numbers.** An open research question; check whether newer results have settled it.
# RE-CHECK	**Whether the coronal heating problem is still considered unsolved.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a campfire graphic where the air gets hotter farther away.
2. Swap the magnetic beat for what the corona looks like in an eclipse.
3. Ask viewers which space mystery to cover next.

Ready to post

TITLE Why the Sun's Atmosphere Is Hotter Than Its Surface

DESCRIPTION The corona tops a million degrees. Nobody has fully explained why.

HASHTAGS #space #sun #science

COVER TEXT Hotter farther away?

PIN Want a script on solar probes next?

The Giant Black Hole at the Heart of Our Galaxy

BEAT TIMELINE (SECONDS) · TOPIC: SAGITTARIUS A*

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	A giant black hole sits at our galaxy's center.	Labelled artist's impression of a galactic center	▸ "OUR BLACK HOLE"	Dramatic
3–10	It's called Sagittarius A star, and it has about four million times the mass of the Sun.	Big number card: about 4,000,000 Suns	Highlight "four million"	Clear
10–18	Don't worry, it's about 26,000 light-years away, and Earth is in no danger from it.	Scale diagram: Sun's position far from the center	‖ Pause after "Don't worry"	Calm
18–27	Astronomers found it by tracking stars near the galactic center for decades, watching them whip around an invisible object.	Animated diagram: star orbits around an empty point	▸ New card per clause	Explaining
27–36	One star swings so close that it reaches thousands of kilometers per second at its nearest point.	Animated icons: star speeding up near the center	Highlight "thousands of km/s"	Amazed
36–45	Only something incredibly massive and compact could pull those orbits so tight, and a black hole fits the evidence.	Simple diagram: tight ellipses around a dark dot	‖ Pause after "compact"	Building
45–54	In 2022, astronomers released the first image of its shadow, a glowing ring much like the one seen in galaxy M87.	Simple graphic: two glowing rings side by side	▸ New card per clause	Wonder
54–60 CTA	Did you know our galaxy has a black hole at its heart? Comment below.	Galaxy icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Sagittarius A* · Encyclopaedia Britannica: Sagittarius A*
SHELF-LIFE	**Evergreen.** A long-measured mass and a historical image; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on stars tracing tight loops around an empty point.
2. Compare its mass with the M87 black hole.
3. Ask viewers whether they knew about it before.

Ready to post

TITLE The Giant Black Hole at the Heart of Our Galaxy

DESCRIPTION Four million Suns' worth of mass, hidden at the Milky Way's center.

HASHTAGS #space #blackhole #milkyway

COVER TEXT A black hole at home

PIN Want the M87 image story next?

The Oldest Light in the Universe Is All Around You

BEAT TIMELINE (SECONDS) · TOPIC: COSMIC MICROWAVE BACKGROUND

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The universe's oldest light is all around you.	Stock clip of a night sky slowly brightening	▸ "THE OLDEST LIGHT"	Wonder
3–10	It's called the cosmic microwave background, a faint glow that fills the whole sky in every direction.	Simple graphic: whole-sky oval map with mottled colors	▸ New card per clause	Clear
10–18	It was released about 380,000 years after the Big Bang, when the universe cooled enough for light to travel freely.	Simple timeline: Big Bang to 380,000 years	Highlight "380,000 years"	Explaining
18–27	That light has traveled for nearly 14 billion years, and expanding space has stretched it into faint microwaves.	Animated icons: wave line stretching longer	‖ Pause after "14 billion years"	Building
27–36	It was first detected in 1965 by two radio astronomers who thought the hiss in their antenna might be pigeon droppings.	Simple graphic: horn antenna icon with a bird icon	▸ New card per clause	Playful
36–45	You've probably heard that old TV static includes it. A small part of that static did come from this ancient glow.	Stock clip of analog TV static	Highlight "a small part"	Clear
45–54	Maps of its tiny temperature differences show the seeds that later grew into galaxies, including ours.	Simple graphic: mottled map zooming into galaxy icons	‖ Pause after "seeds"	Wonder
54–60 CTA	Does it amaze you that old static held Big Bang light? Comment below.	TV icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Cosmic Microwave Background · Encyclopaedia Britannica: cosmic microwave background
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Most of old TV static was light from the Big Bang. **Documented:** Only a small part of analog TV static, often quoted as about one percent, came from the cosmic background.

Make it yours

3 swaps so it's your video

1. Open on old TV static with a small highlighted fraction.
2. Swap the pigeon beat for what the CMB map colors mean.
3. Ask viewers if they remember analog TV static.

Ready to post

TITLE The Oldest Light in the Universe Is All Around You

DESCRIPTION A faint glow from the early universe still fills the whole sky.

HASHTAGS #space #bigbang #cosmology

COVER TEXT The oldest light

PIN Want a script on how old the universe is?

People Have Lived in Orbit Nonstop Since 2000

BEAT TIMELINE (SECONDS) · TOPIC: INTERNATIONAL SPACE STATION

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	People have lived in orbit nonstop since 2000.	Labelled artist's impression of a space station over Earth	▸ "NONSTOP SINCE 2000"	Curious
3–10	The International Space Station has had a crew on board continuously since November 2000.	Simple timeline starting at November 2000	Highlight "continuously"	Clear
10–18	As of the mid-2020s, that's **more than two decades** without a single day empty.	Counter icon showing years, still running	‖ Pause after "mid-2020s"	Proud
18–27	It orbits about 400 kilometers up and circles Earth roughly every 90 minutes, so the crew sees about 16 sunrises a day.	Animated diagram: orbit loop with a 90-minute timer	▸ New card per clause	Amazed
27–36	It's about the size of a football field, built piece by piece in orbit by several space agencies working together.	Scale diagram: station outline over a football field	Highlight "football field"	Explaining
36–45	On a clear night, you can see it without a telescope: a bright, steady dot gliding across the sky in a few minutes.	Stock clip of night sky, moving dot graphic	▸ New card per clause	Wonder
45–54	It won't last forever. Its partners have announced plans to retire it in the coming years.	Simple timeline with a fading end	‖ Pause after "forever"	Calm
54–60 CTA	Have you ever spotted the station? Tell me where you saw it.	Station icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA: International Space Station facts and figures · Encyclopaedia Britannica: International Space Station
SHELF-LIFE	**Re-check numbers.** Crew status and retirement plans change; the voiceover hedges the record.
# RE-CHECK	**Whether the ISS is still crewed, and its current retirement plans.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a calendar flipping from 2000 to today.
2. Swap the size beat for what crews do on a normal day.
3. Ask viewers to share a photo of the station passing over.

Ready to post

TITLE People Have Lived in Orbit Nonstop Since 2000

DESCRIPTION The space station has had a crew every day since 2000.

HASHTAGS #space #iss #astronaut

COVER TEXT Nonstop since 2000

PIN Want a guide to spotting the station?

The Longest Single Trip to Space: About 437 Days

BEAT TIMELINE (SECONDS) · TOPIC: LONGEST SINGLE SPACEFLIGHT

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	The longest single spaceflight lasted about 437 days.	Big number card: about 437 days	▸ "437 DAYS"	Curious
3–10	As of the mid-2020s, that record still belongs to Valeri Polyakov, a Russian doctor and cosmonaut.	Simple graphic: record badge with a calendar	Highlight "still belongs"	Clear
10–18	He flew to the Mir space station in 1994 and came home in 1995, after more than fourteen months in orbit.	Simple timeline: 1994 to 1995	▸ New card per clause	Explaining
18–27	As a doctor, part of his mission was to study what a long stay does to the human body and mind.	Animated icons: stethoscope, brain and heart icons	‖ Pause after "As a doctor"	Calm
27–36	That question matters for Mars. A round trip there could take well over a year, even with a short stay.	Simple diagram: Earth to Mars path with a calendar	Highlight "well over a year"	Building
36–45	After landing, he chose to walk a short distance from the capsule instead of being carried.	Animated icons: footsteps leaving a capsule outline	▸ New card per clause	Proud
45–54	Long stays still cause bone loss, muscle loss and eye changes, so researchers keep studying crews on long missions.	Simple graphic: three health icons with arrows	‖ Pause after "eye changes"	Serious
54–60 CTA	Could you spend fourteen months in a small space station? Comment below.	Calendar icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	Encyclopaedia Britannica: Valery Polyakov, and Mir (space station) · NASA: Human Research Program, long-duration spaceflight
SHELF-LIFE	**Re-check numbers.** Records can be broken; the voiceover dates the record to the mid-2020s.
# RE-CHECK	**Longest single spaceflight record: who holds it and how many days.**
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a day counter racing to 437.
2. Swap the walking beat for what a Mars crew would need.
3. Ask viewers how long they'd last in a small station.

Ready to post

TITLE The Longest Single Trip to Space: About 437 Days

DESCRIPTION A doctor spent over fourteen months on Mir to study long stays.

HASHTAGS #space #astronaut #mars

COVER TEXT 437 days in orbit

PIN Want a script on the health risks of Mars trips?

Why Stars Twinkle but Planets Shine Steady

BEAT TIMELINE (SECONDS) · TOPIC: TWINKLING

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Stars twinkle. Planets mostly shine steadily.	Stock clip of a twinkling night sky	▸ "WHY THE TWINKLE?"	Curious
3–10	The stars themselves aren't flickering. The twinkle happens in Earth's air, not in space.	Simple diagram: star, atmosphere layer, eye	▸ New card per sentence	Firm
10–18	Our atmosphere is full of moving pockets of warmer and cooler air that bend light slightly as it passes through.	Animated icons: wavy air layers bending a light ray	Highlight "bend light"	Explaining
18–27	A star is so far away that it's basically a single point of light, so every tiny bend makes it jump and flicker.	Simple graphic: single dot jittering	‖ Pause after "single point"	Clear
27–36	Planets are much closer, so they appear as tiny disks. The flickers from different parts of the disk average out.	Simple diagram: small disk made of many points	▸ New card per sentence	Explaining
36–45	That's a handy trick: if a bright dot shines steadily, there's a good chance it's a planet.	Simple graphic: steady dot labelled planet	Highlight "shines steadily"	Light
45–54	Stars near the horizon twinkle most, because their light crosses more air. Astronauts above the air see steady stars.	Simple diagram: long low light path vs short high path	‖ Pause after "twinkle most"	Calm
54–60 CTA	Have you used the twinkle test to find a planet? Tell me below.	Star icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Space Place: Why Do Stars Twinkle? · Encyclopaedia Britannica: scintillation (astronomy)
SHELF-LIFE	Evergreen. Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Stars twinkle because they flicker or pulse. **Documented:** The twinkling comes from moving air in Earth's atmosphere bending their light.

Make it yours

3 swaps so it's your video

1. Open on a pool-floor light pattern as the comparison.
2. Swap the horizon beat for how telescopes correct twinkling.
3. Ask viewers to try the twinkle test tonight.

Ready to post

TITLE Why Stars Twinkle but Planets Shine Steady

DESCRIPTION The twinkle is Earth's air, not the star.

HASHTAGS #space #stargazing #stars

COVER TEXT Why stars twinkle

PIN Did the twinkle test work for you?

Neptune Was Found With Math

BEAT TIMELINE (SECONDS) · TOPIC: NEPTUNE'S DISCOVERY

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Neptune was predicted with math before it was spotted.	Animated icons: equation lines turning into a blue planet	▸ "FOUND WITH MATH"	Curious
3–10	In the 1800s, astronomers noticed Uranus wasn't moving quite where its orbit said it should be.	Simple diagram: predicted path vs actual path of Uranus	Highlight "wasn't moving"	Clear
10–18	Something unseen seemed to be tugging on it. A French mathematician, Urbain Le Verrier, worked out where that something must be.	Animated icons: invisible tug arrow, pencil and paper	▸ New card per sentence	Explaining
18–27	In 1846, he sent his prediction to the Berlin Observatory, and astronomers there found the planet the very first night they looked.	Simple sky chart with an X marking the spot	‖ Pause after "1846"	Building
27–36	It was within about one degree of the predicted spot, a tiny patch of sky.	Simple graphic: small circle around the X	Highlight "one degree"	Amazed
36–45	An English mathematician, John Couch Adams, had made similar calculations, and both are now often credited.	Simple graphic: two pencil icons side by side	▸ New card per clause	Calm
45–54	Neptune is so far out that one year there lasts about 165 Earth years. It finished its first full orbit since discovery in 2011.	Animated diagram: one full orbit loop, 1846 to 2011	‖ Pause after "165 Earth years"	Wonder
54–60 CTA	Would you rather find a planet with math or a telescope? Comment below.	Math and telescope icons + comment bubble	Hold last card 1.5 s	Playful

VERIFY WITH	NASA Science: Neptune Facts · Encyclopaedia Britannica: Neptune (planet), discovery section
SHELF-LIFE	Evergreen. A historical discovery; nothing here changes year to year.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on a hand-written equation fading into a blue planet.
2. Swap the Adams beat for Neptune's fast winds.
3. Ask viewers whether they'd rather do the math or the observing.

Ready to post

TITLE Neptune Was Found With Math

DESCRIPTION A planet predicted on paper, then found the first night they looked.

HASHTAGS #space #neptune #history

COVER TEXT Found with math

PIN Want a script on how Pluto was found?

Our Moon May Have Formed From a Giant Collision

BEAT TIMELINE (SECONDS) · TOPIC: GIANT-IMPACT HYPOTHESIS

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Our Moon may have formed from a giant collision.	Labelled artist's impression of two young worlds colliding	▸ "BORN FROM A CRASH?"	Dramatic
3–10	The leading idea is that a Mars-sized body slammed into the young Earth about 4.5 billion years ago.	Scale diagram: young Earth and a Mars-sized body	Highlight "Mars-sized"	Clear
10–18	Scientists often call that body Theia. The impact would have blasted molten rock and vapor into orbit around Earth.	Animated icons: debris ring forming around Earth	▸ New card per sentence	Building
18–27	Over time, that debris clumped together and formed the Moon, possibly much closer to Earth than it is today.	Animated diagram: ring clumping into a Moon	‖ Pause after "clumped together"	Wonder
27–36	Evidence: Moon rocks brought back by Apollo crews are strikingly similar to Earth's rocks in their chemistry.	Simple graphic: two rock icons with matching labels	Highlight "strikingly similar"	Explaining
36–45	The Moon also has a small iron core, which fits the idea that it formed mostly from rocky outer layers.	Simple diagram: Moon cut-away with a small core	▸ New card per clause	Clear
45–54	Some details are still debated, like how much of the Moon came from Theia, but the collision idea leads the field.	Simple graphic: question mark beside a leading badge	‖ Pause after "debated"	Calm
54–60 CTA	Does a collision origin make the Moon more fascinating to you? Comment below.	Moon icon + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Science: Moon Formation · Encyclopaedia Britannica: Moon, origin and evolution section
SHELF-LIFE	**Evergreen.** A long-standing leading hypothesis; the voiceover presents it as an idea, not proven fact.
MYTH CHECK	No common myth.

Make it yours

3 swaps so it's your video

1. Open on two planet icons colliding in slow motion.
2. Swap the iron-core beat for how the young Moon looked from Earth.
3. Ask viewers which origin story surprised them most.

Ready to post

TITLE Our Moon May Have Formed From a Giant Collision

DESCRIPTION The leading idea: a Mars-sized body hit the young Earth.

HASHTAGS #space #moon #science

COVER TEXT Born from a crash?

PIN Want a script on how the Moon is drifting away?

Auroras Happen at Both Ends of the Earth

BEAT TIMELINE (SECONDS) · TOPIC: AURORAS

0 3 10 18 27 36 45 54 60

TIME (S)	VOICEOVER	VISUAL	CAPTION / PAUSE	TONE
0–3 HOOK	Auroras happen at both ends of the Earth.	Stock clip of a green aurora	▸ "BOTH POLES"	Wonder
3–10	Many people know the northern lights, but the southern lights are just as real, around the Antarctic region.	Simple globe graphic: rings at both poles	▸ New card per clause	Clear
10–18	They start with the Sun, which constantly sends out a stream of charged particles called the solar wind.	Animated icons: particles streaming from the Sun	Highlight "solar wind"	Explaining
18–27	Earth's magnetic field steers many of those particles toward the polar regions, where they slam into our upper atmosphere.	Simple diagram: field lines guiding particles to the poles	‖ Pause after "polar regions"	Building
27–36	When they hit oxygen and nitrogen, those gases glow. Oxygen often gives green, and sometimes red at higher altitudes.	Simple graphic: color key, green and red labelled oxygen	▸ New card per sentence	Clear
36–45	Nitrogen can add blue and purple tints, which is why the edges of some displays look violet.	Simple graphic: blue and purple swatches labelled nitrogen	Highlight "blue and purple"	Light
45–54	During big solar storms, auroras can reach much farther from the poles, and phone cameras often catch colors our eyes miss.	Animated icons: phone camera showing brighter colors	‖ Pause after "solar storms"	Wonder
54–60 CTA	Northern or southern lights: which would you travel to see? Comment below.	Aurora icons + comment bubble	Hold last card 1.5 s	Warm

VERIFY WITH	NASA Space Place: What Is an Aurora? · Encyclopaedia Britannica: aurora
SHELF-LIFE	**Evergreen.** Settled science; nothing here changes year to year.
MYTH CHECK	**Popular version:** Auroras only happen near the North Pole. **Documented:** They happen around both poles; the southern lights circle the Antarctic region.

Make it yours

3 swaps so it's your video

1. Open on a split screen: northern and southern auroras.
2. Swap the color beat for how solar storms push auroras south.
3. Ask viewers whether they've seen one on a phone camera.

Ready to post

TITLE Auroras Happen at Both Ends of the Earth

DESCRIPTION Northern lights have a southern twin. Here's what makes them glow.

HASHTAGS #space #aurora #northernlights

COVER TEXT Southern lights exist too

PIN Want a script on the Sun's 11-year cycle?

Turn a script into a video in 6 steps

A generic workflow for any image library, any text-to-speech tool or your own voice, any editor with captions and any simple graphics app.

1. **Personalize and check**
   Apply at least one “Make it yours” swap, look up the core claim in the “Verify with” reference, and confirm any figure named on a “Re-check” line. Read the script aloud once with a timer.

2. **Record the voiceover**
   Paste the Voiceover column into any text-to-speech tool, or record yourself at about 150 words per minute. Leave the pauses from the Caption / pause column.

3. **Gather visuals you have rights to**
   Public-domain agency images (check each credit line), labelled artist’s impressions, simple scale diagrams and animated icons from any simple graphics app, plus licensed stock clips of the night sky. No film or TV clips.

4. **Cut beat by beat**
   Use the Time (s) column as your edit list: one cut or overlay change at each beat boundary. The beat timeline on each page shows where they fall.

5. **Add captions**
   Big captions, 1–3 words per card, made in any editor with captions. Highlight the words named in the Caption / pause column and hold the final question on screen for about 1.5 s.

6. **Post and pin**
   Use the “Ready to post” box: title, description, hashtags and cover text. Turn on any disclosure the platform requires, and pin the suggested comment.

Batch tip: the CSV edit list

Open `script_beats.csv` in Google Sheets or Excel and filter by `script_id`. Each row is one beat with start and end seconds, voiceover, visual, caption and tone: an edit list to keep open next to your editor.

Make a series

Pair a script with a hook of the same theme for a follow-up video, or use the Myth busted hooks to open a series on space myths and what is actually documented.

FAQ

Which themes are covered?

8 themes: Planets of the Solar System; Moons & our Moon; Sun & stars; Black holes & extreme objects; Galaxies & the scale of the universe; Missions & exploration history; Humans in space (body, food, sleep, suits); Sky myths & everyday astronomy. Each has at least 7 scripts.

How many hooks are there?

150 hooks: 10 hook types × 15 hooks each (H001–H150). 8 of the 10 types are space-only structures (Scale it down to Cosmic extremes); Myth busted and Would you? complete the set.

What do the Shelf-life tags mean?

Evergreen = physics and long-settled facts that won't change. Re-check numbers = the script contains a count, record or mission status that can change; the figure is hedged in the voiceover and named on the script's "Re-check" line, so confirm it before posting.

What is NOT covered?

- Conspiracy content (fake Moon landing, flat Earth, "NASA is hiding..."), except where a Myth check debunks it
- Astrology or horoscopes
- UFO or alien-sighting claims
- Doomsday or "asteroid will hit on [date]" predictions
- Live news or upcoming launch dates
- Rocket or astronaut career advice
- Analysis of real living people (astronauts or company founders) as subjects
- Video clips or images (visual cues only)
- Non-English text

Can I add my own hooks?

Yes. hooks.csv is a plain spreadsheet you can extend with your own rows. There is no generator.

Does it batch-generate a week of scripts?

No. It is a library of 60 finished scripts, not a generator or an app. 60 scripts = 2 months at 1 per day, or 3 weeks at 3 per day.

Does the same story come in 30, 45 and 60 s versions?

No. Each story is written at one length only: 20 at 30 s, 20 at 45 s and 20 at 60 s. The longest script is 60 s.

Is this affiliated with NASA or any space agency?

No. Faceless Creator Kit is not affiliated with NASA, ESA, JPL or any space agency. Agency pages are named only as places to check a claim. We are not astronomers; check each claim against the named reference before posting.

License

Simple version: use the scripts in your own content as much as you like; don’t resell or share the pack.

You can

- Use any hook or script, as written or edited, in your own videos, Shorts, Reels, posts and captions.
- Use them for personal or commercial purposes, including your own monetized channels, on any number of your own channels.
- Edit, shorten, extend, translate or combine them, and read them with your own voice or any text-to-speech voice.
- Import the CSV files into your own tools, sheets and editing workflows, and add your own rows.

You can’t

- Resell, share, give away or upload this pack, the PDFs or the CSV files, in whole or in part.
- Publish the scripts or hooks as a script product: a script pack, template, prompt pack, course, e-book or any other text product.

Good to know

- One purchase covers one creator, or one channel team working on the same channels.
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